



Conceptual Level | Entity-Relationship Model

Privacy and Personal Information on Cloud Services

1.4.2. Team and Individual Contributions

- Each team must maintain a Git-based repository (e.g., GitLab, GitHub, ...)⁷ for the project. The repository must be associated with the individual accounts of all team members. The repository will be used to document the development of the project and to make individual contributions traceable. The repository should be used throughout the project rather than only immediately before a submission deadline. The instructor and/or marker may consider the repository's commit history, pull requests, issues, documentation, and other project records when assess individual contributions. The number of commits alone will not be used as a measure of contribution. The quality, significance, and nature of each member's contributions will also be considered.
- All team members are expected to make substantive contributions, and no member should be limited solely to coordination or project management.

Author
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⁷ Web-based applications and cloud services may collect, use and/or disclose your personal information and store or access that information outside of Canada. Personal information includes, but not limited to, identifiable individual's information such as your name or your email address. At UVic, GitLab at gitlab.uvic.ca is highly recommended for coursework, research and development projects. For more information, see <https://www.uvic.ca/systems/services/pages/gitlab.php>. At your team's discretion, you may use GitHub to store information about your project, but we do not require nor recommend storing any personal identifying information. Also, we do not require public repo, but it should be accessible to instructors and markers for evaluation purposes.



Milestone 2: Background and Related Work

Project

3

1. Project

1

Definition and
Team Formation

[2. Background and
Related Work](#)

1

2. Background and Related Work

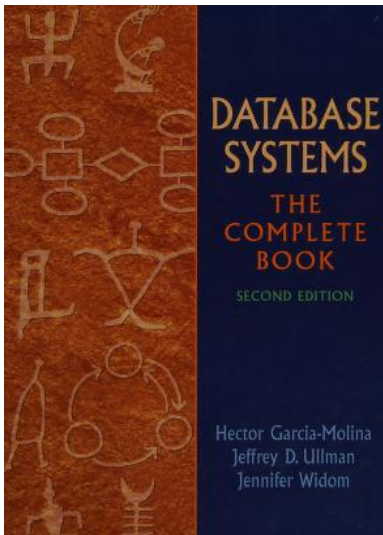


Assignment

Due October 8 at 5:00 AM

Please see the attached manual and sample submission. The sample is **for illustration only and is not a complete or authoritative example**. Please follow the manual for the actual requirements.

4 Outline: W02B: Thursday



CH04 (2nd Ed.)

I) Entity

II) Attribute

III) Relationship

III) Extended ER (EER)

5

Entity-Relationship Model (ER)

1. Identify real-world **Entities**, **Attributes**, **Relationships**
2. Create graphical schema: ER Diagram (ERD)

A picture is worth a thousand words!

Standard → all data engineers all over the world understand ERD!

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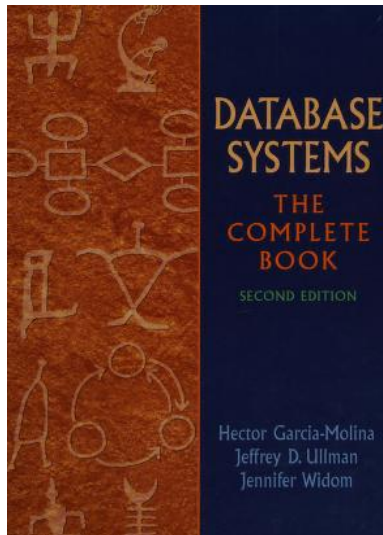
Entity-Relationship Model: Entity Set (E)

A set of entities of the same type that share the same properties

Movie	$=\{e_1:\text{The Birds}, e_2:\text{Rosemary's Baby}, e_3:\text{L.A. Confidential}, \dots\}$
Director	$=\{e_4:\text{Alfred Hitchcock}, e_5:\text{Roman Polanski}, e_6:\text{James Cameron}, \dots\}$
Actor	$=\{e_7:\text{Tippi Hedren}, e_8:\text{Mia Farrow}, e_9:\text{John Cassavetes}, \dots\}$
Instructor	$=\{e_{10}:\text{Kovvy}, e_{11}:\text{Hosseini}, \dots\}$
Singer	$=\{e_{12}:\text{Bruno Mars}, e_{13}:\text{Taylor Swift}, \dots\}$
Novel	$=\{e_{14}:\text{19Q4}, e_{15}:\text{Brothers Karamazov}, \dots\}$

Entity (~~Set~~)

Outline: W02B



CH04 (2nd Ed.)

I) Entity

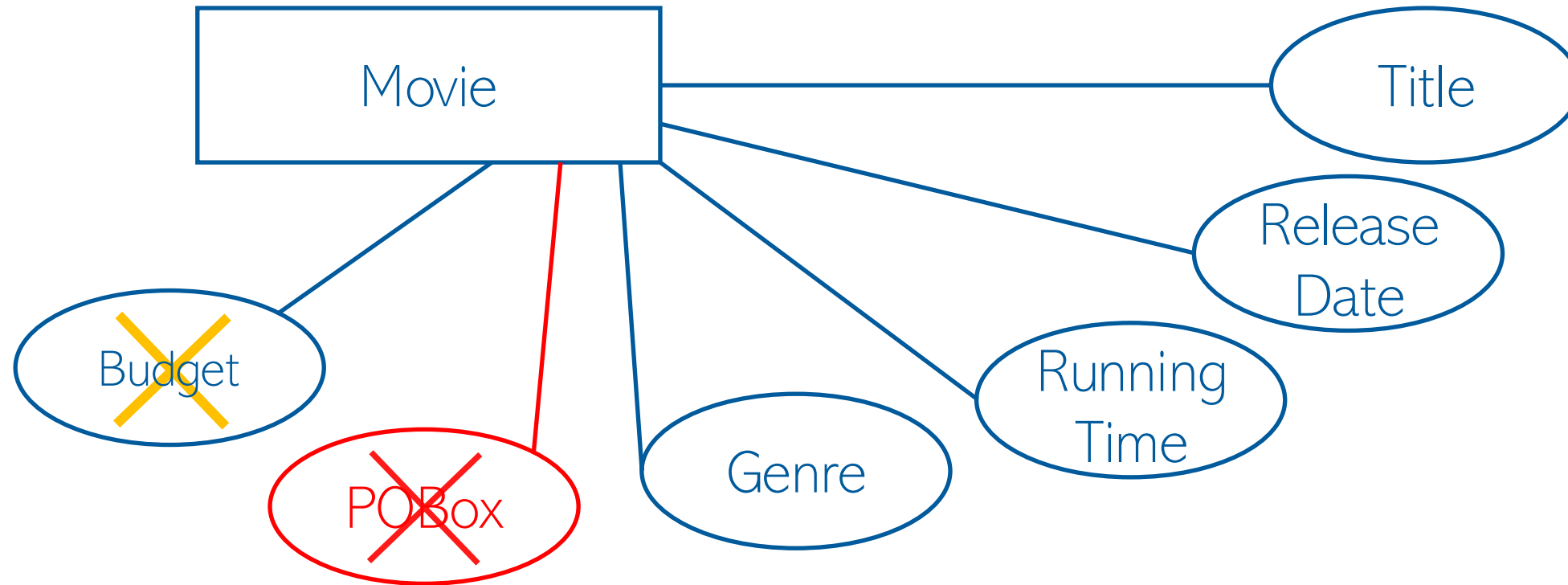
II) Attribute

III) Relationship

IIII) Extended ER (EER)

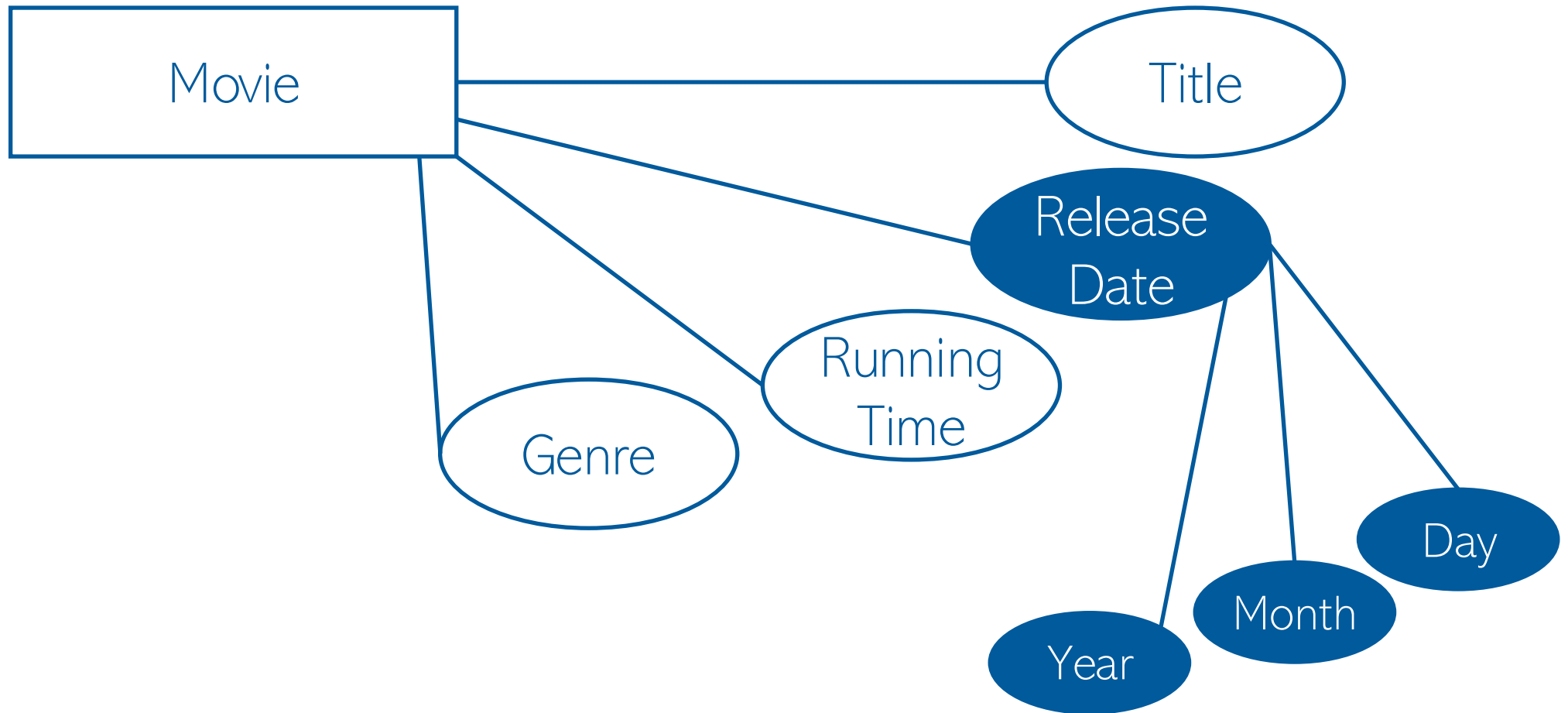
8

Attribute: Faithfulness



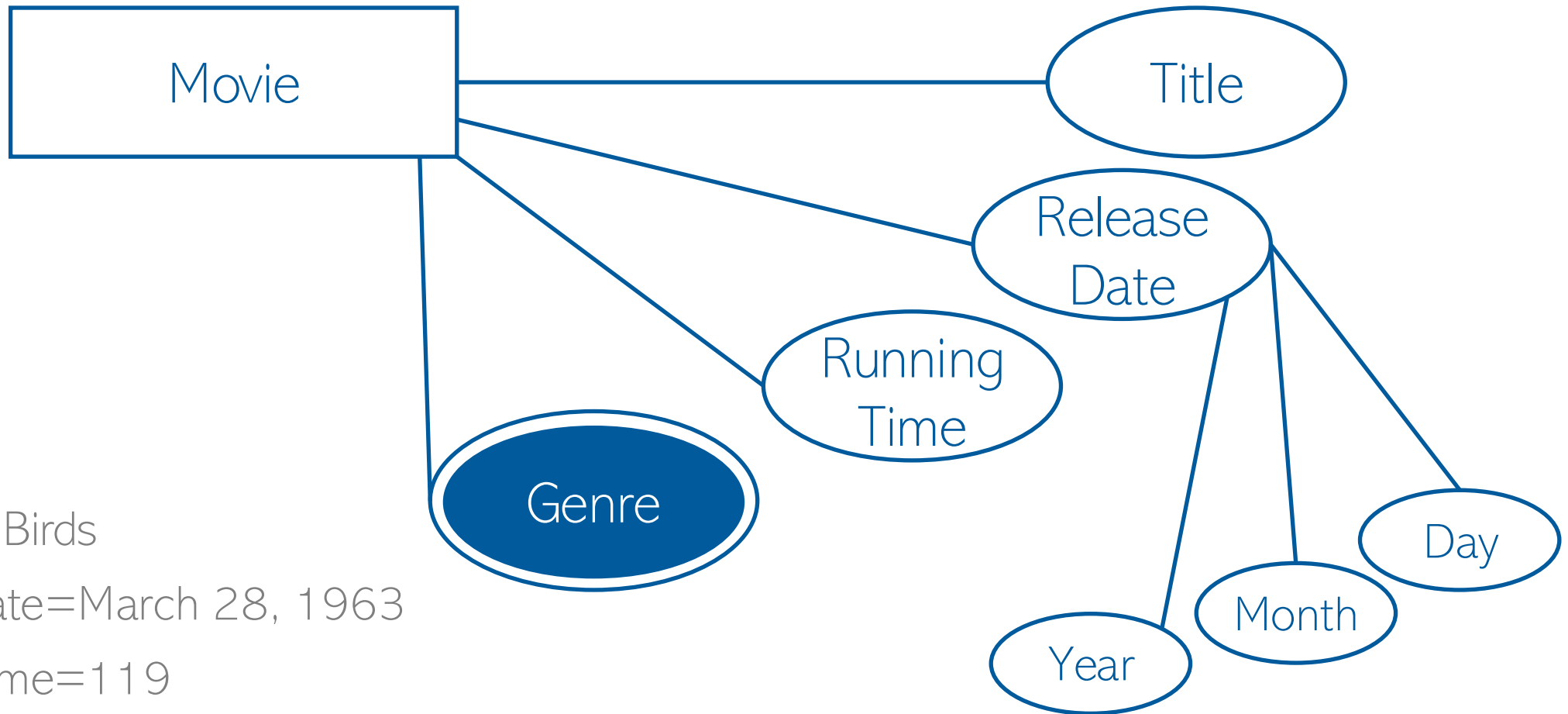
9

Attribute: Composite



10

Attribute: Multivalued



e_1 .Title=The Birds

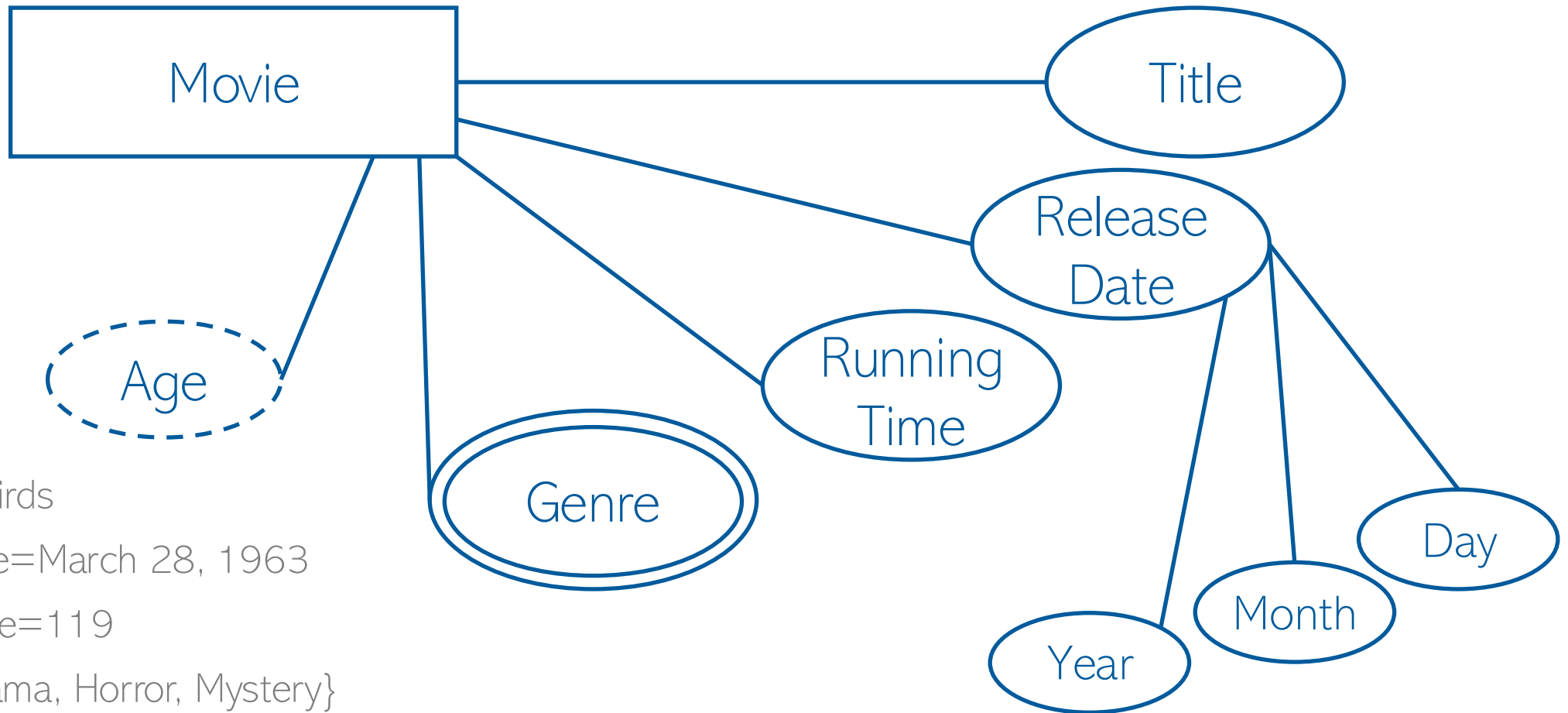
e_1 .ReleaseDate=March 28, 1963

e_1 .RunningTime=119

e_1 .Genre={Drama, Horror, Mystery}

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Attribute: Derived



e_1 .Title=The Birds

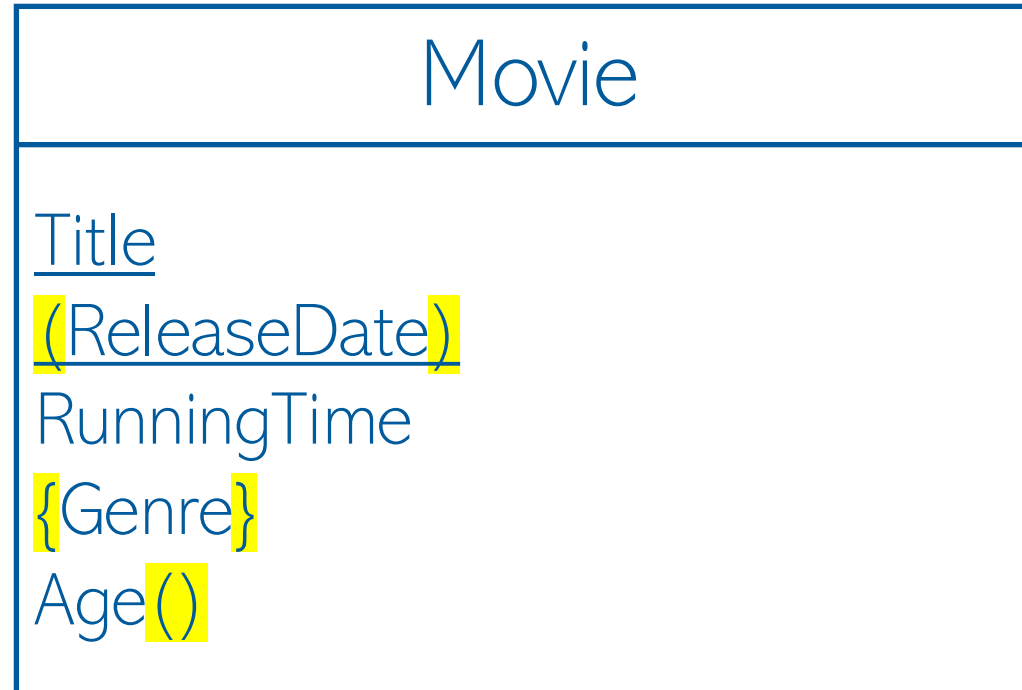
e_1 .ReleaseDate=March 28, 1963

e_1 .RunningTime=119

e_1 .Genre={Drama, Horror, Mystery}

e_1 .Age=YEAR(e_1 .ReleaseDate-NOW)

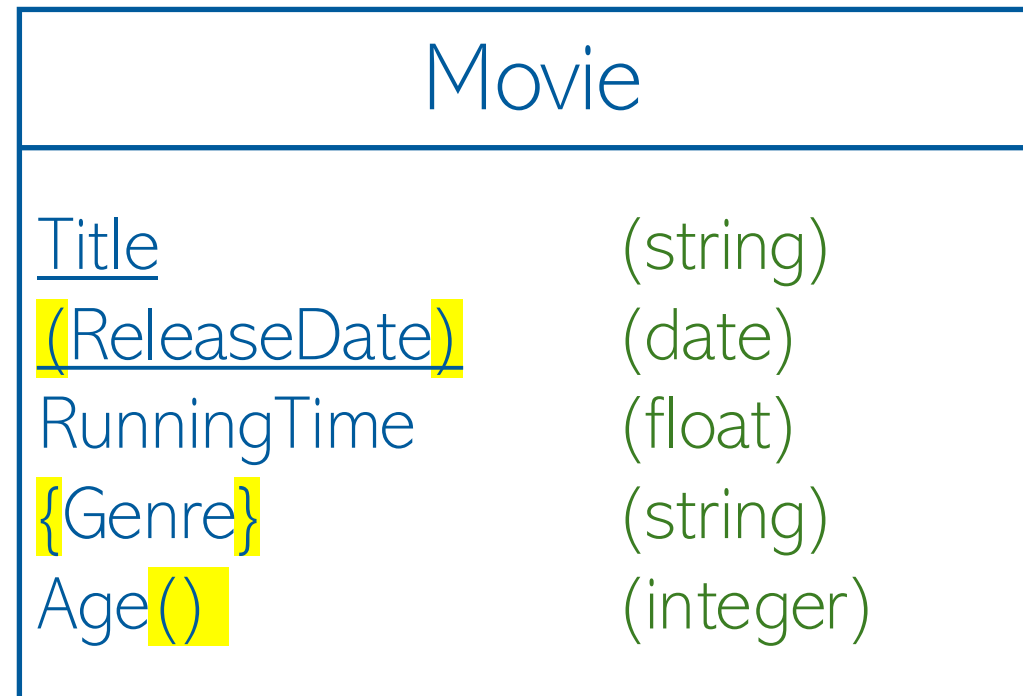
Entity + Attribute: Alternative Notation



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Entity + Attribute: Alternative Notation: Data Type

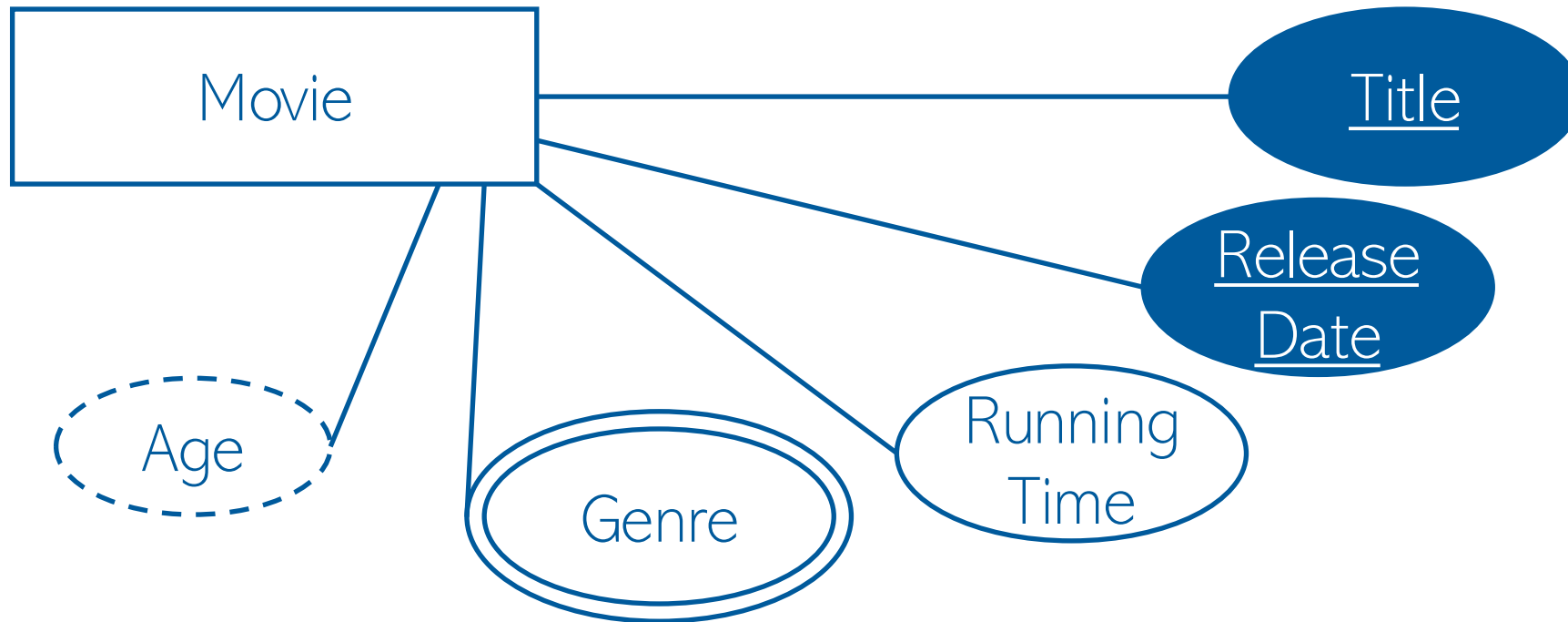
Standard E/R model does not have!



14

Attribute: Key

One or a set of attributes uniquely identify an entity in entity set



Rosemary's Baby (1968) by Roman Polanski

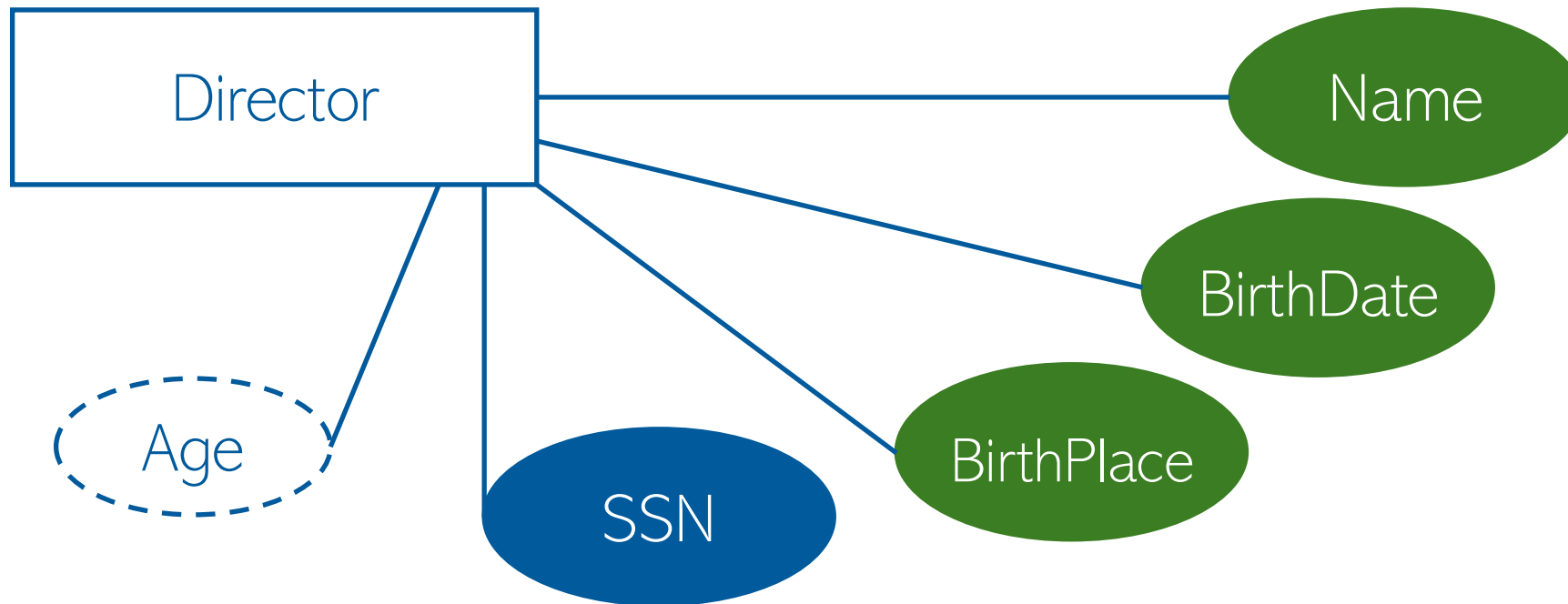
Rosemary's Baby (2014) by Agnieszka Holland (TV Mini-Series)

Rosemary's Baby (2015) by Dane Kessel

15

Attribute: Multiple Keys

Standard E/R model does not have!



$K_1 = \{\text{Name, BirthDate, BirthPlace}\}$

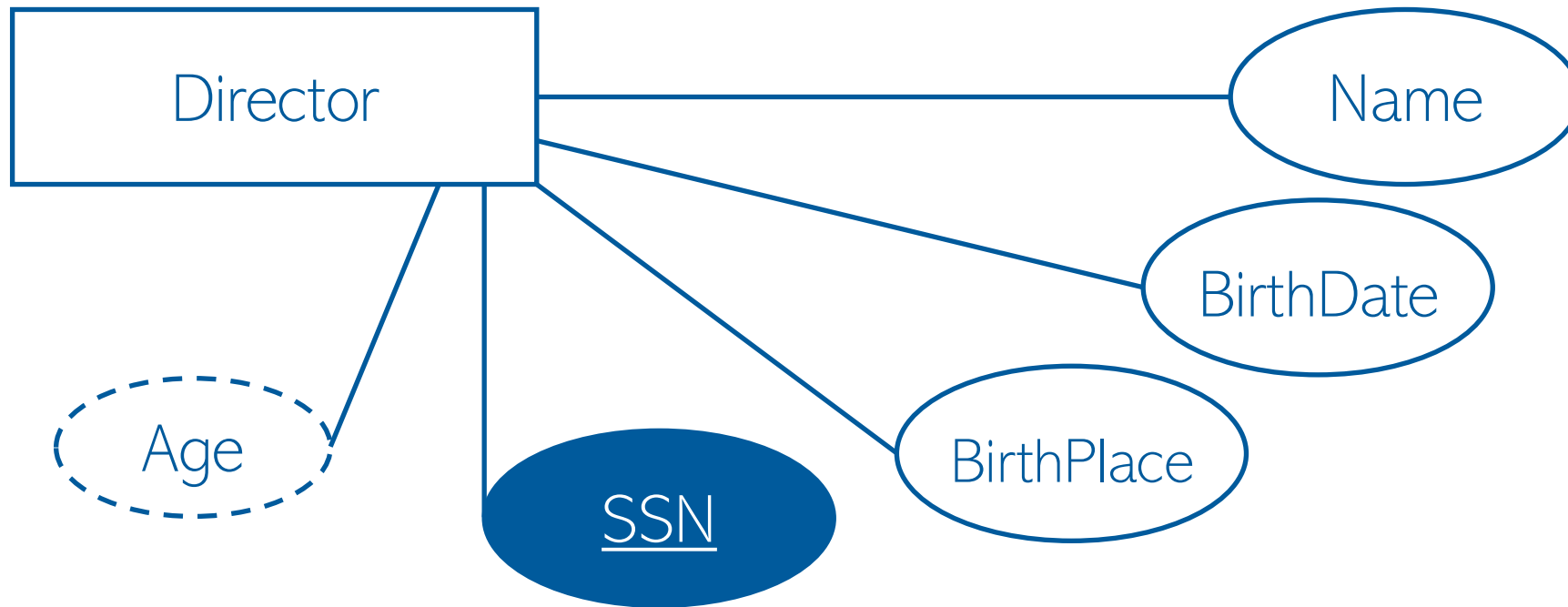
$K_2 = \{\text{SSN}\}$

Choose?

16

Attribute: Multiple Keys

Standard E/R model does not have!



$K_1 = \{\text{Name, BirthDate, BirthPlace}\}$

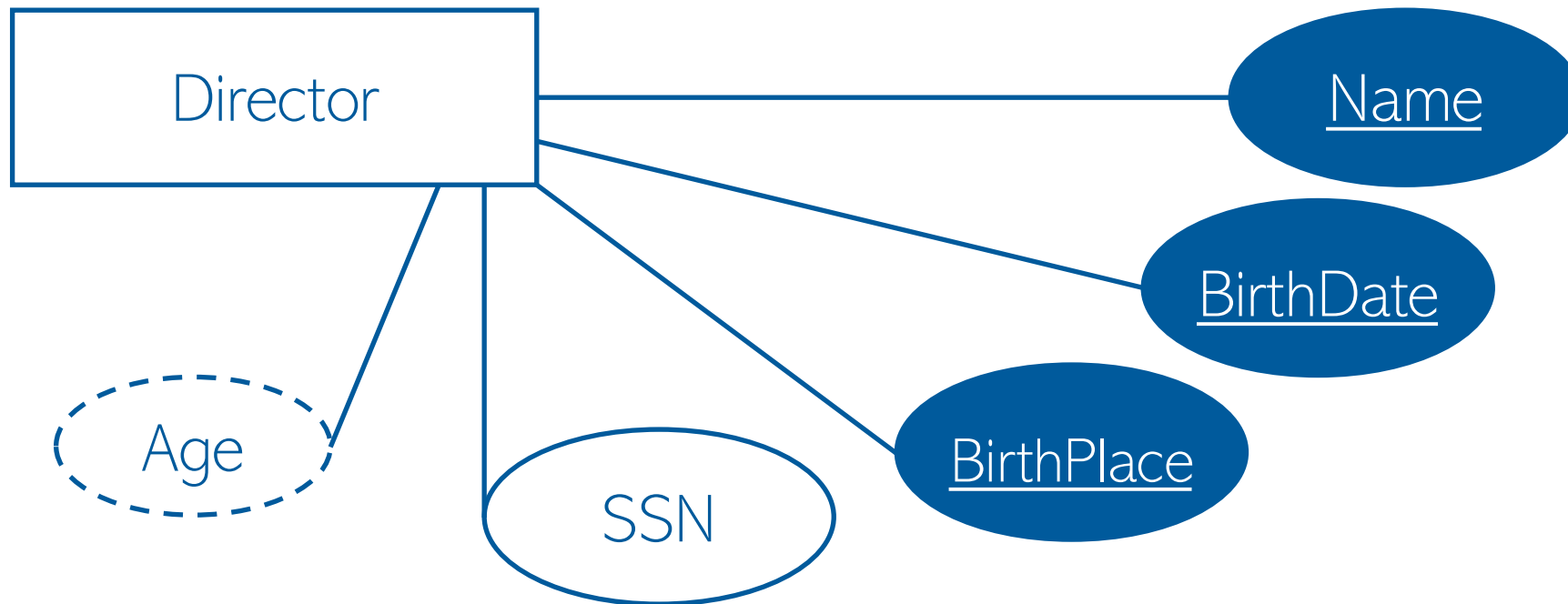
$K_2 = \{\text{SSN}\}$

Choose? Since simplicity matters, we may opt for K_2 as the Primary Key (PK)

17

Attribute: Multiple Keys

Standard E/R model does not have!

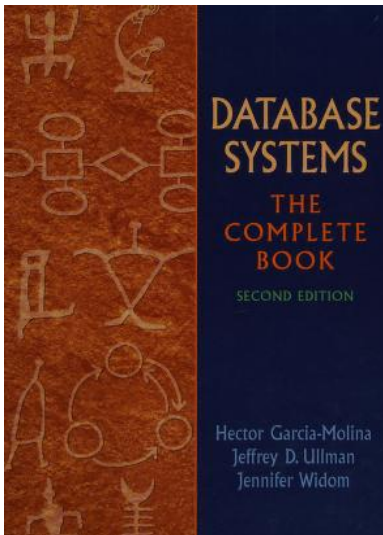


$K_1 = \{\text{Name, BirthDate, BirthPlace}\}$

$K_2 = \{\text{SSN}\}$

Choose? Since not all directors has SSN, we must select K_1 as the Primary Key

Outline: W02B



CH04 (2nd Ed.)

I) Entity

II) Attribute

III) Relationship

IIII) Extended ER (EER)

Relationship Set (R)

$r = (e_1, e_2) = (\{\text{The Birds, March-28-1963}\}, \{\text{Alfred Hitchcock, August-13-1899}\})$

$r' = (e'_1, e'_2) = (\{\text{Rosemary's Baby, June-12-1968}\}, \{\text{Roman Polanski, August-18-1933}\})$

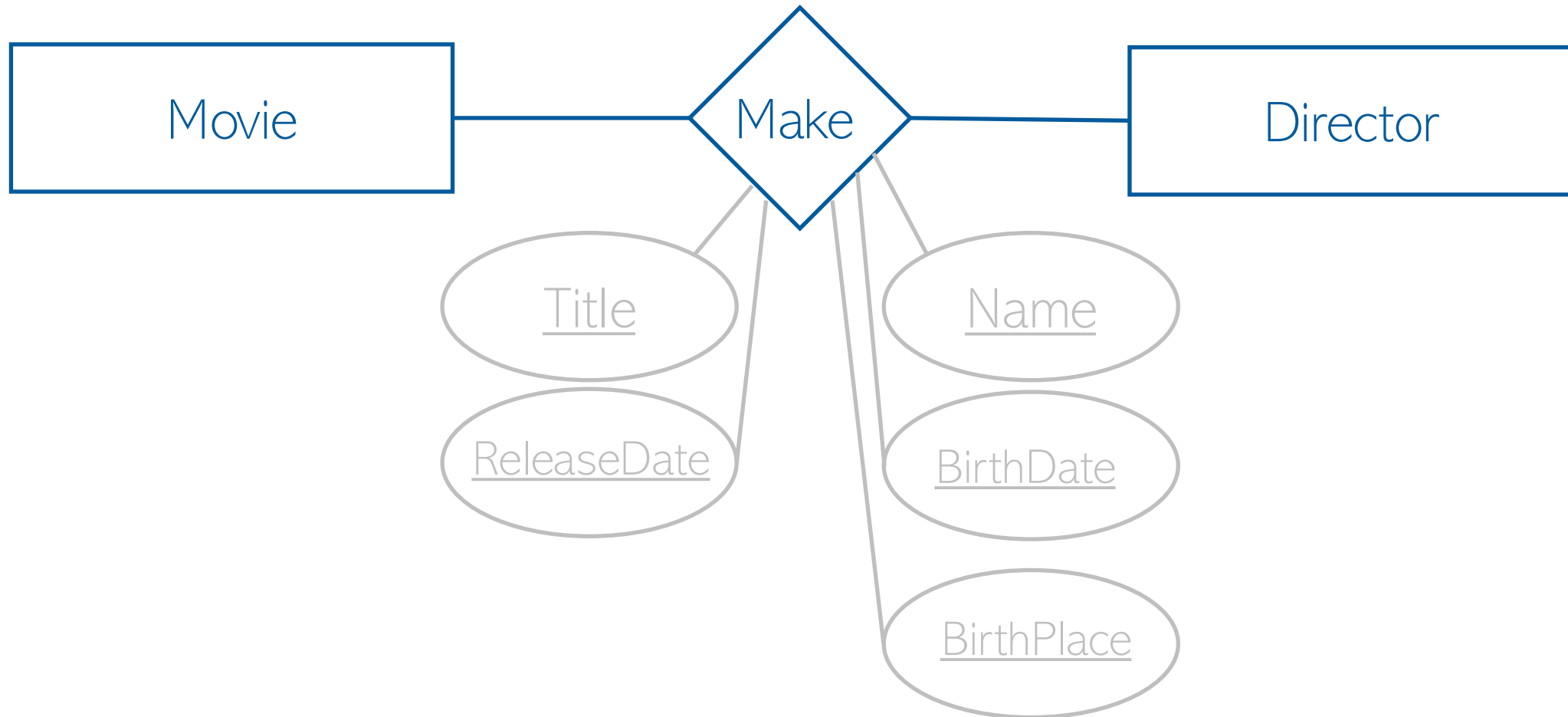
...

Binary Relationships

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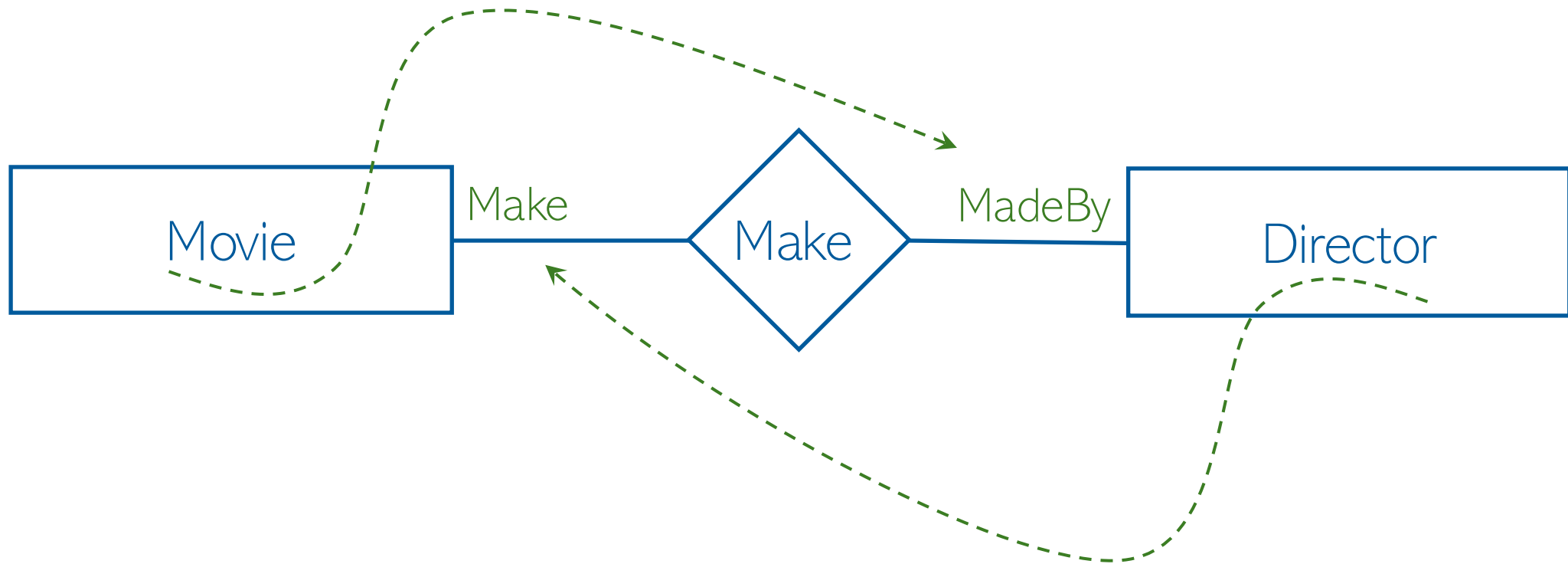
Relationship Set (R)

We do not show the keys from participating entity sets.



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Relationship Set (R): Roles (when not clear)



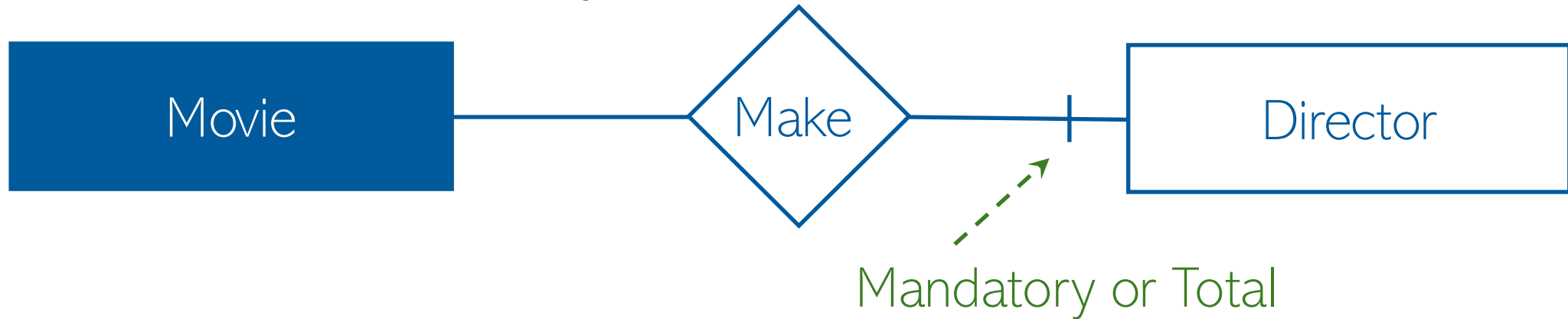
22

Relationship Set (R): Participation (Ordinality)

A Movie MUST participate in Make relationship? **Yes.**

A Movie MUST have Director? **Yes.**

Is there a Movie in the system without Director? **No!**



Movie's participation in Make relationship with Director is mandatory (total).

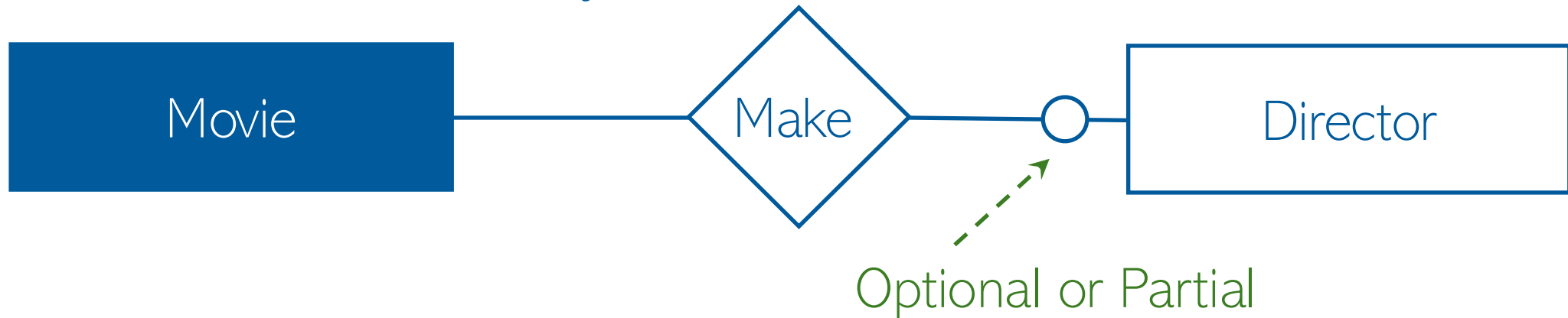
23

Relationship Set (R): Participation (Ordinality)

A Movie MUST participate in Make relationship? No!

A Movie MUST have Director? No!

Is there a Movie in the system without Director? Yes.



Movie's participation in Make relationship with Director is optional (paartial).

Relationship Set (R): Participation (Ordinality)

Minimum number of times an entity in an entity set can be associated with another entity in the related entity set.

Either	0: Optional, Partial
Or	1: Mandatory, Total

Relationship Set (R): Multiplicity (Cardinality)

Maximum number of times an entity in an entity set can be associated with another entity in the related entity set.

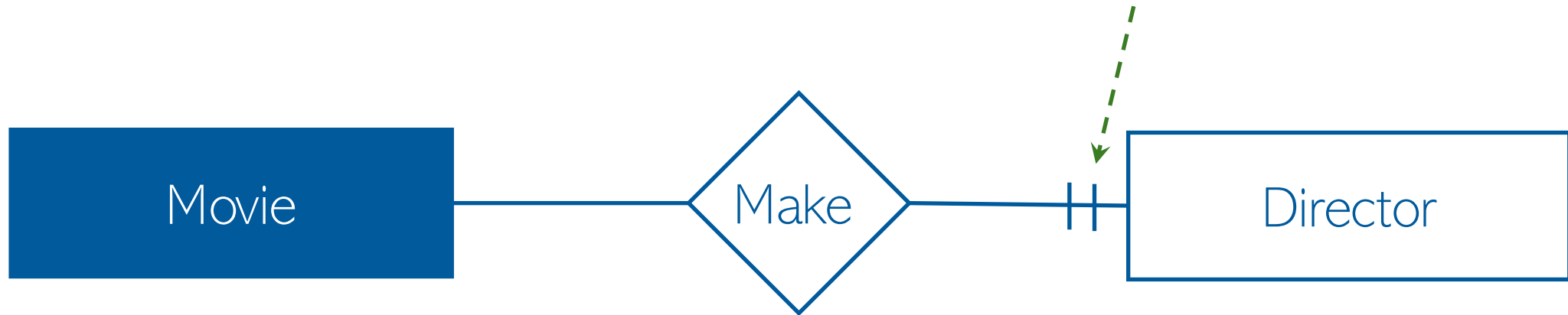
Either 1

Or Many (more than 1)

26

Relationship Set (R): Multiplicity (Cardinality)

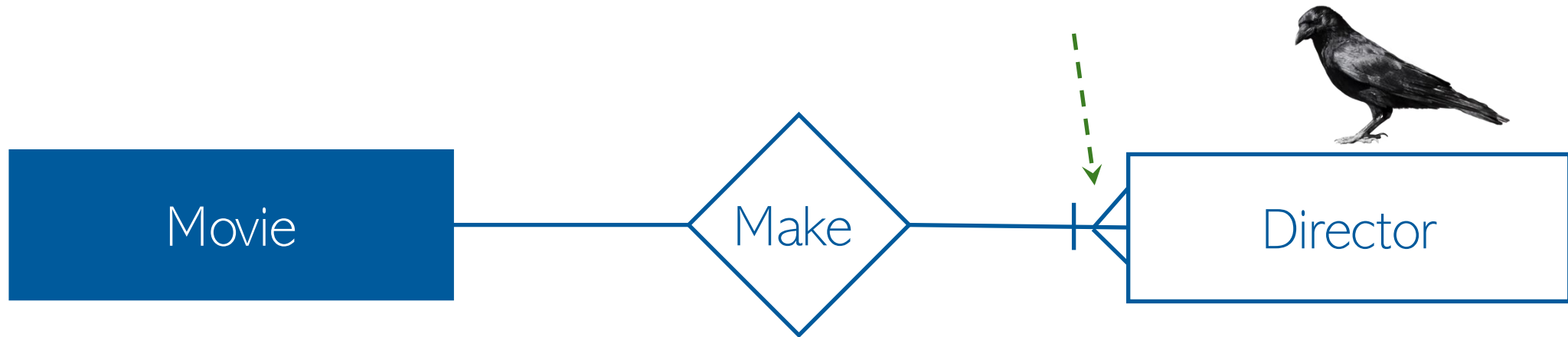
A Movie participate in Make relationship with 1 Director at max



27

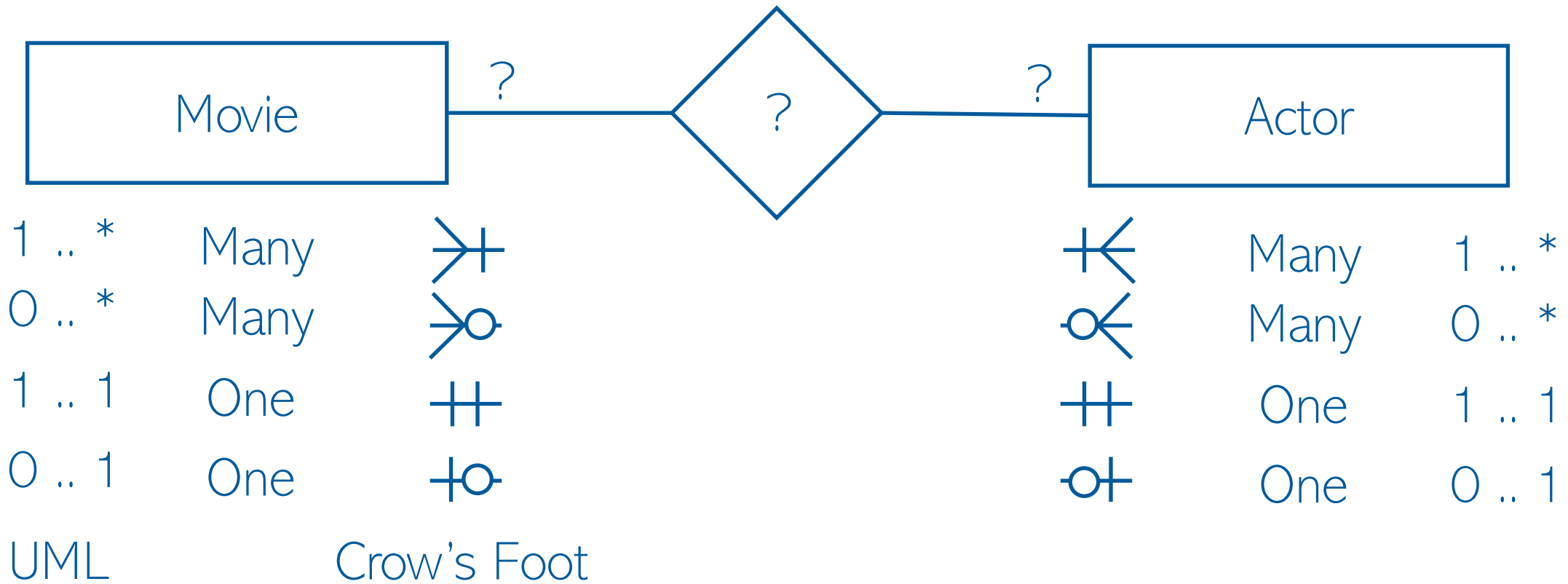
Relationship Set (R): Multiplicity (Cardinality)

A Movie participate in Make relationship with many Director at max



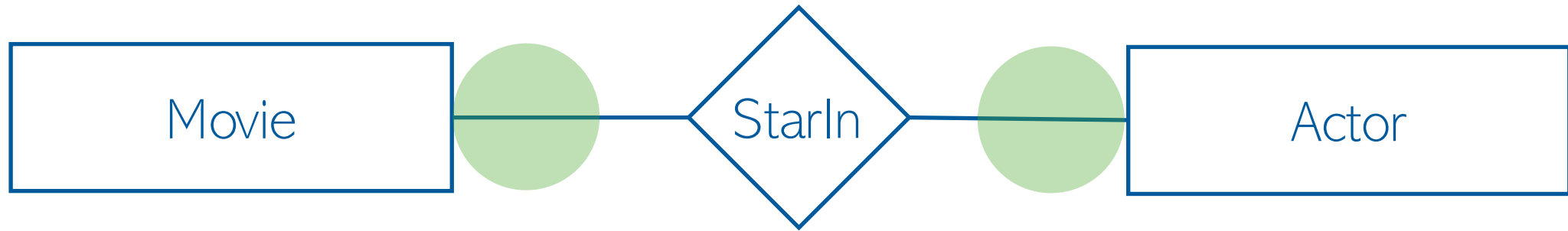
28

Relationship Set (R): Multiplicity (Cardinality)

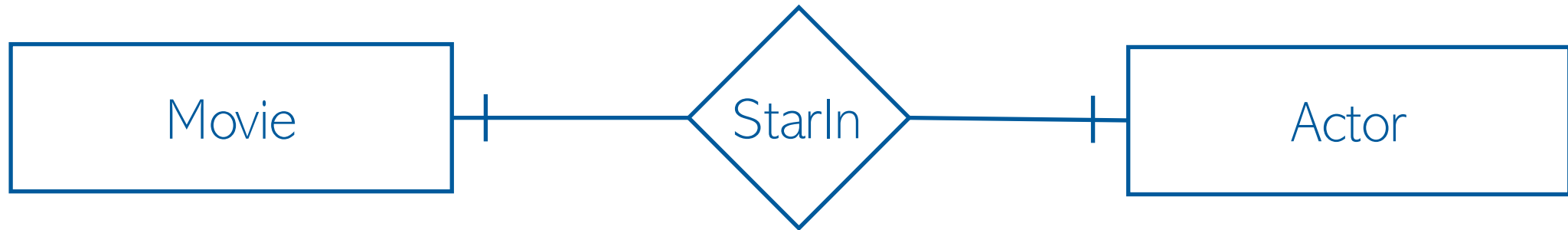


29

Relationship Set (R): Participation and Multiplicity



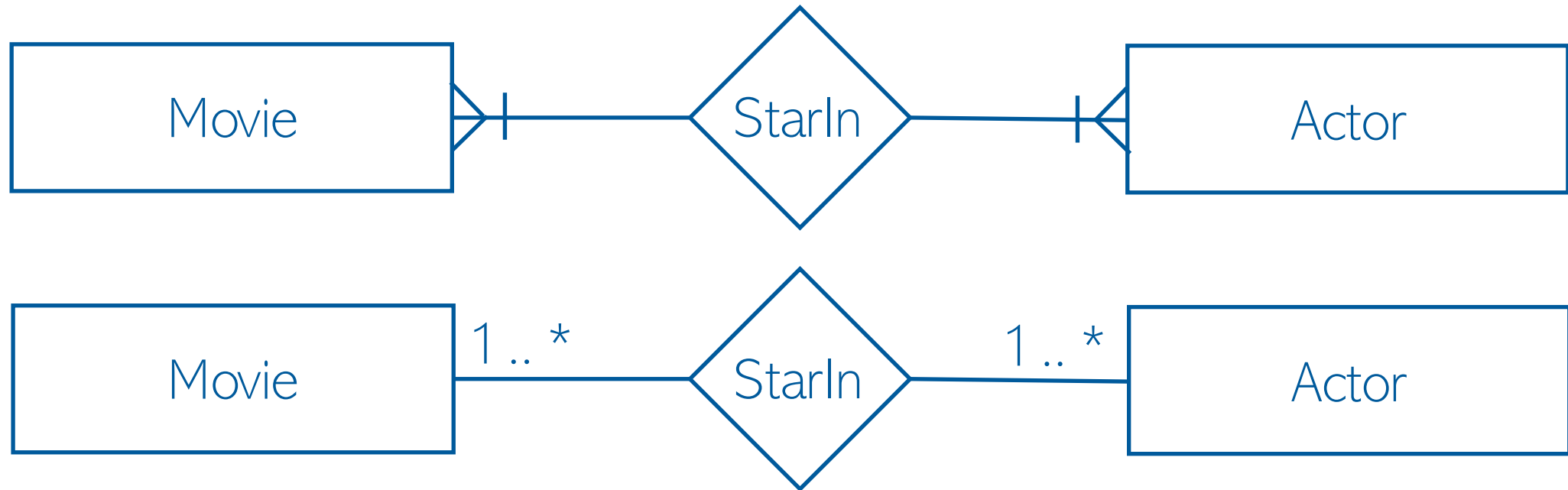
Relationship Set (R): Participation and Multiplicity



A Movie must have at least 1 Actor entity (Mandatory).
An Actor must have at least 1 Movie entity (Mandatory).

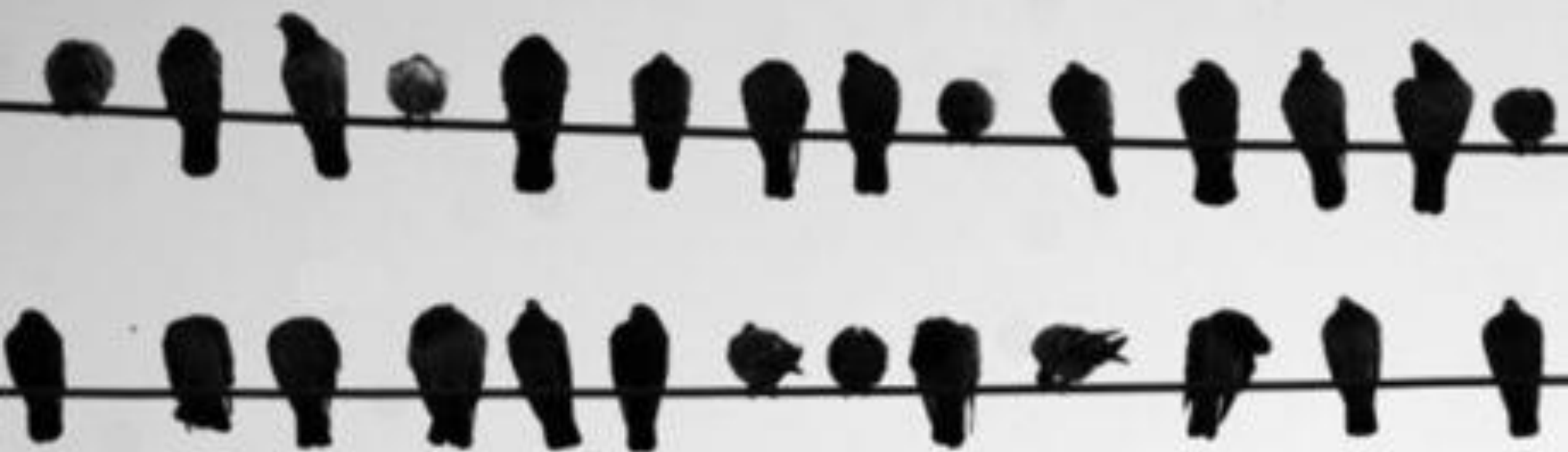
31

Relationship Set (R): Participation and Multiplicity

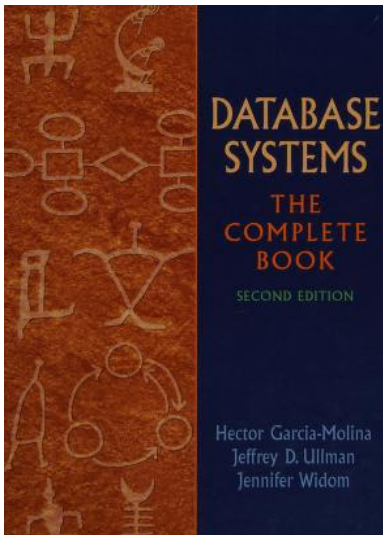


A Movie may have more than 1 Actor entity (Many).
An Actor may have more than 1 Movie entity (Many).

ALFRED HITCHCOCK S



Outline: W03A



CH04 (2nd Ed.)

I) Entity

II) Attribute

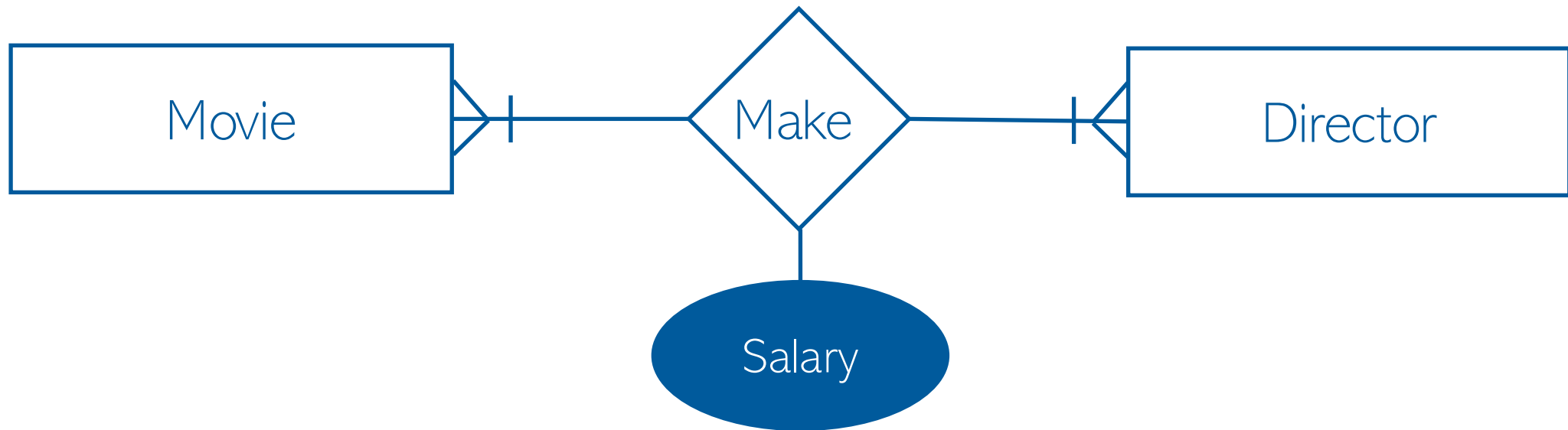
III) Relationship

IIII) Extended ER (EER)

34

Relationship Set (R): **Attribute**

Exist if and only if there is a relationship



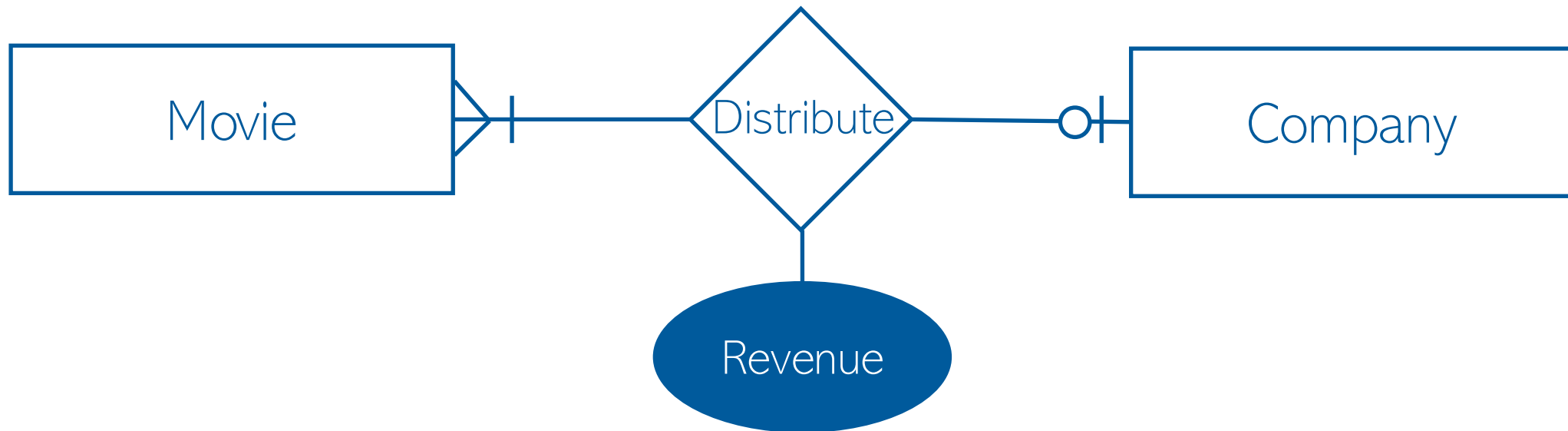
E.g., Salary between a Movie entity and a Director entity.

'Hitchcock' was paid '\$X' to make 'The Birds'.

35

Relationship Set (R): Attribute

Exist if and only if there is a relationship

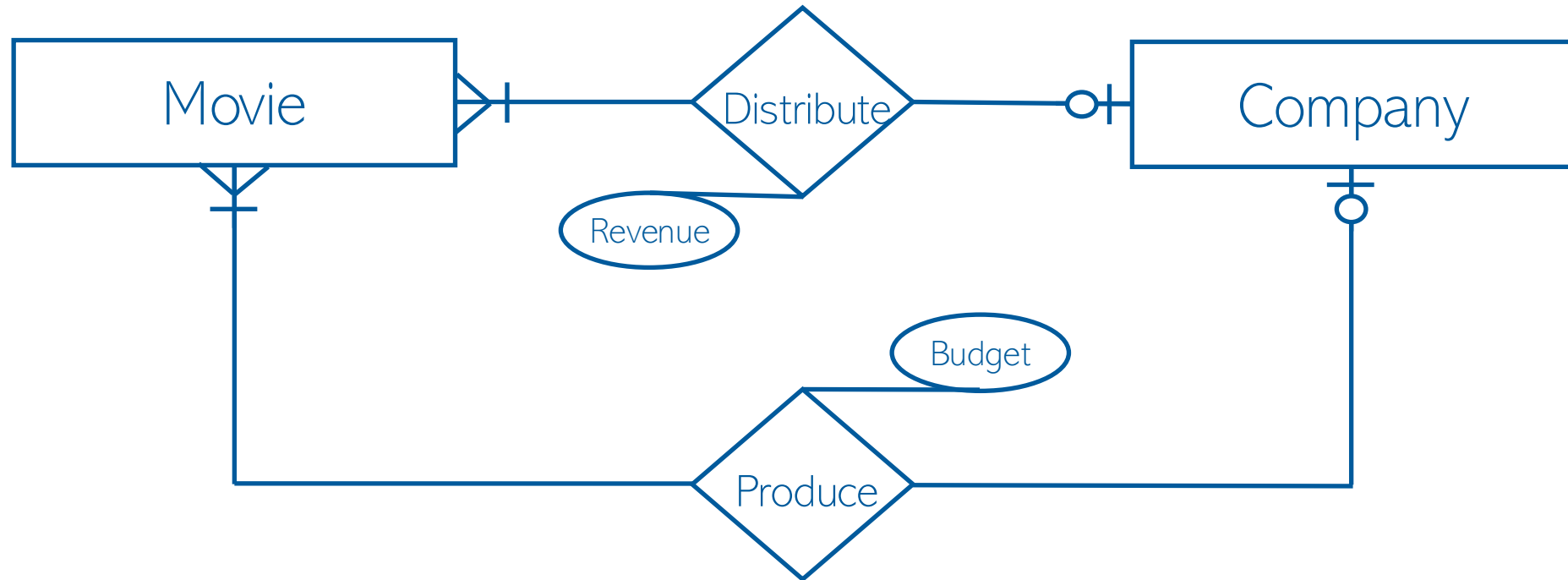


E.g., Revenue for a Movie when distributed by a Company entity.
'Universal Pictures' make '\$3.3 Million' for distributing 'The Birds'.

36

Relationship Set (R): Multiple

A pair of entities of different sets have different types of relationship



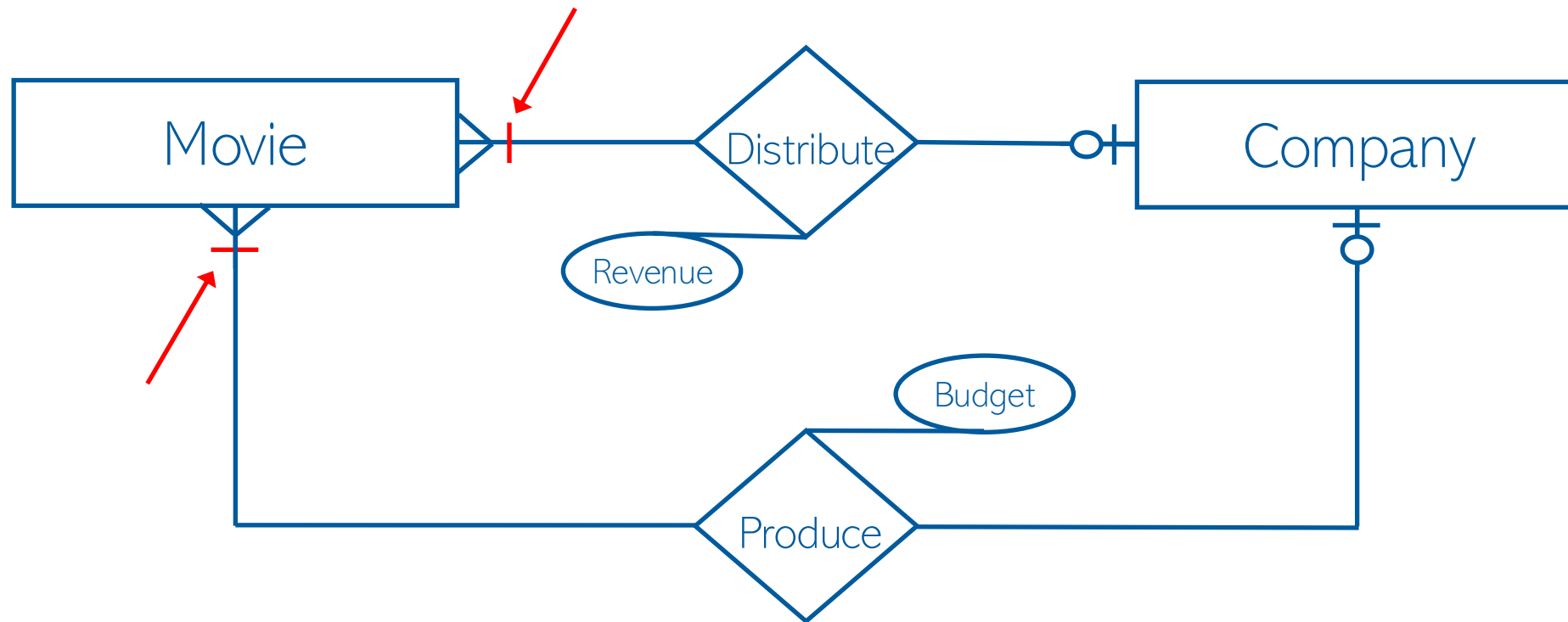
E.g., A Movie is Produced by A Company entity. It may be distributed by the same, or different Company entity.

'The Birds' was made by 'AIH Inc.' and distributed by 'Universal Pictures'.

37

Relationship Set (R): Multiple

A pair of entities of different sets have different types of relationship

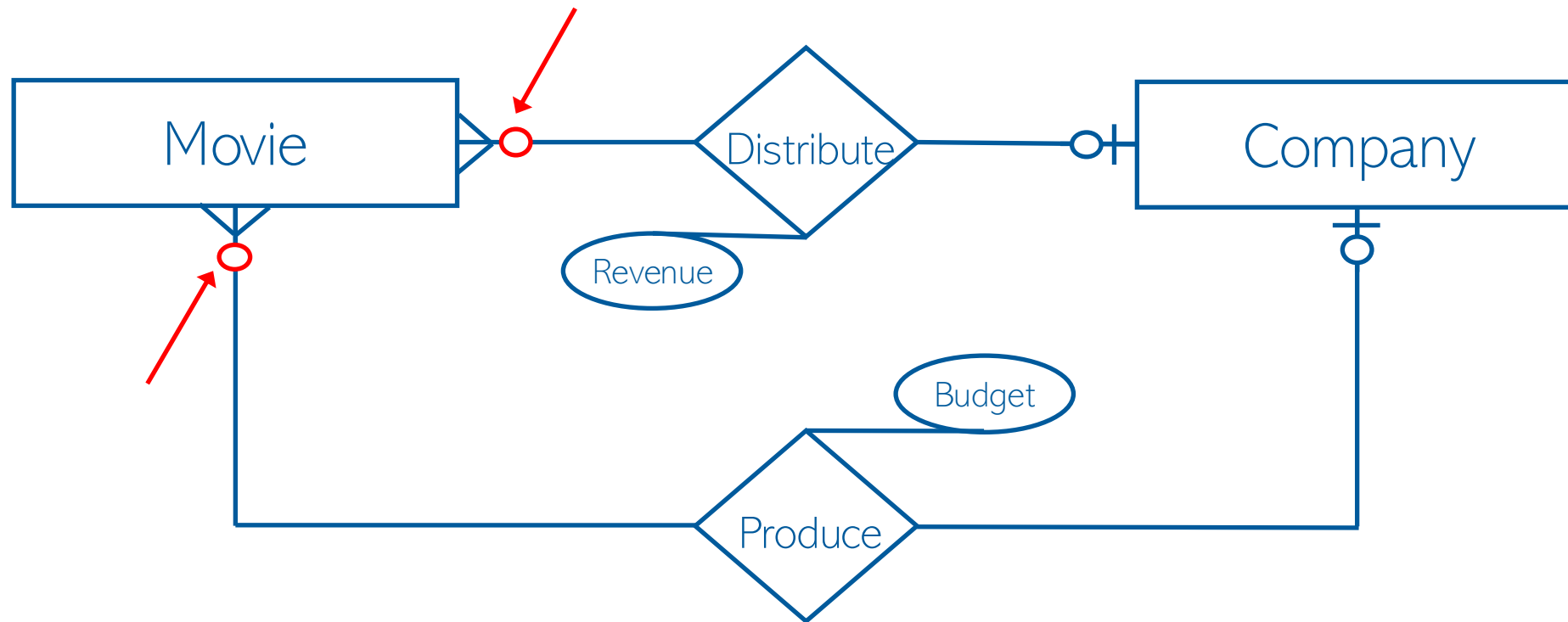


A Company entity **must act as both** Produce and Distribute entity.
'The Birds' was made by 'AIH Inc.' and distributed by 'Universal Pictures'.

38

Relationship Set (R): Multiple

A pair of entities of different sets have different types of relationship



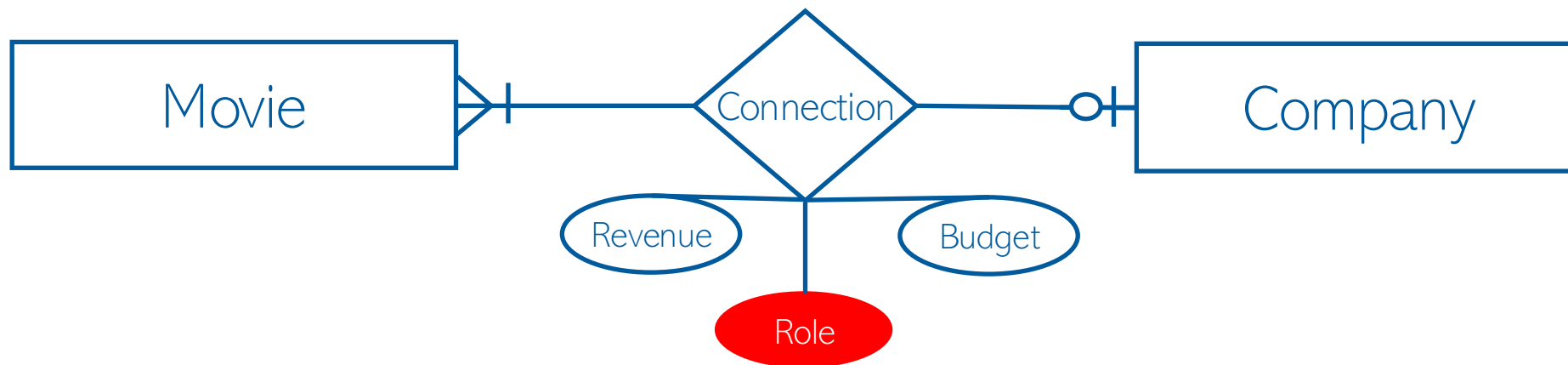
A Company entity **may neither** Produce **nor** Distribute entity.

'The Birds' was made by 'AII Inc.' and distributed by 'Universal Pictures'.

39

Relationship Set (R): Multiple

A pair of entities of different sets have different types of relationship



Not explicit about the relationship:

What is the Connection relationship?

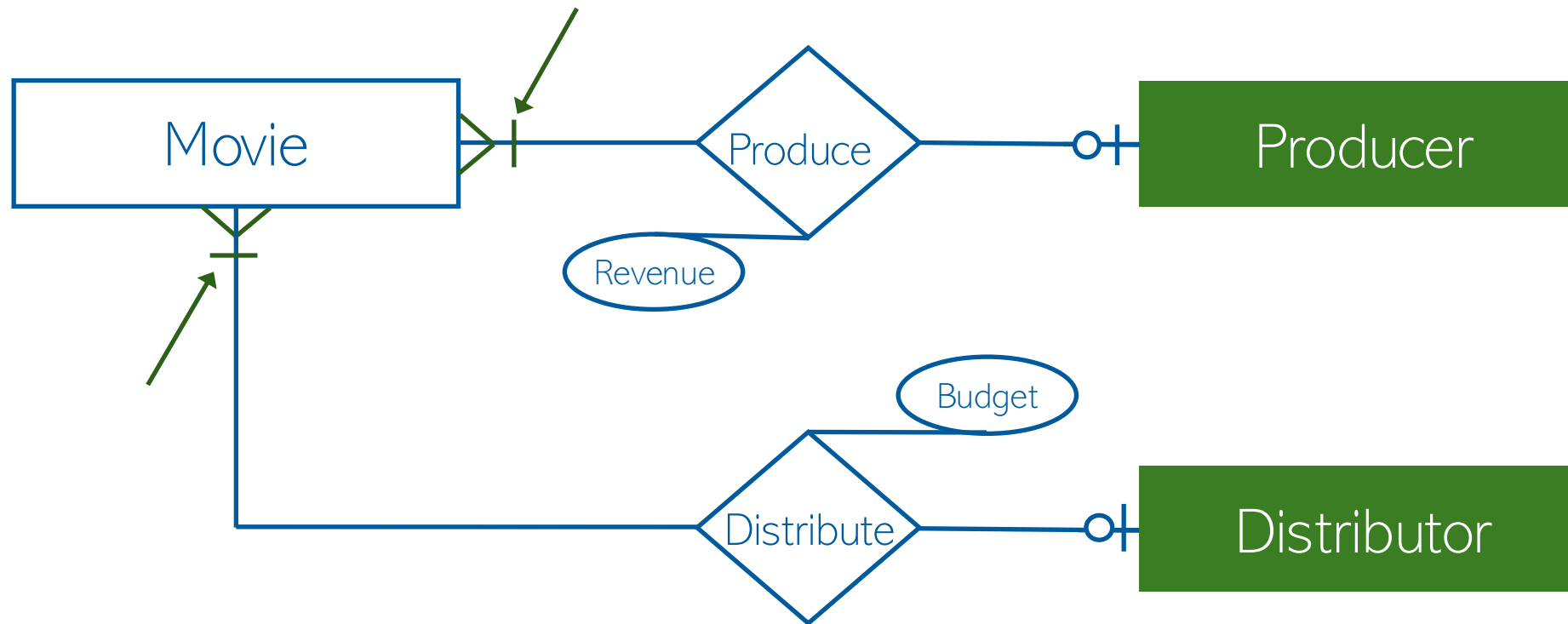
Mixing the mutual attributes of relationships:

If Distributor then Revenue, Else If Producer then Budget.

40

Relationship Set (R): Multiple

A pair of entities of different sets have different types of relationship



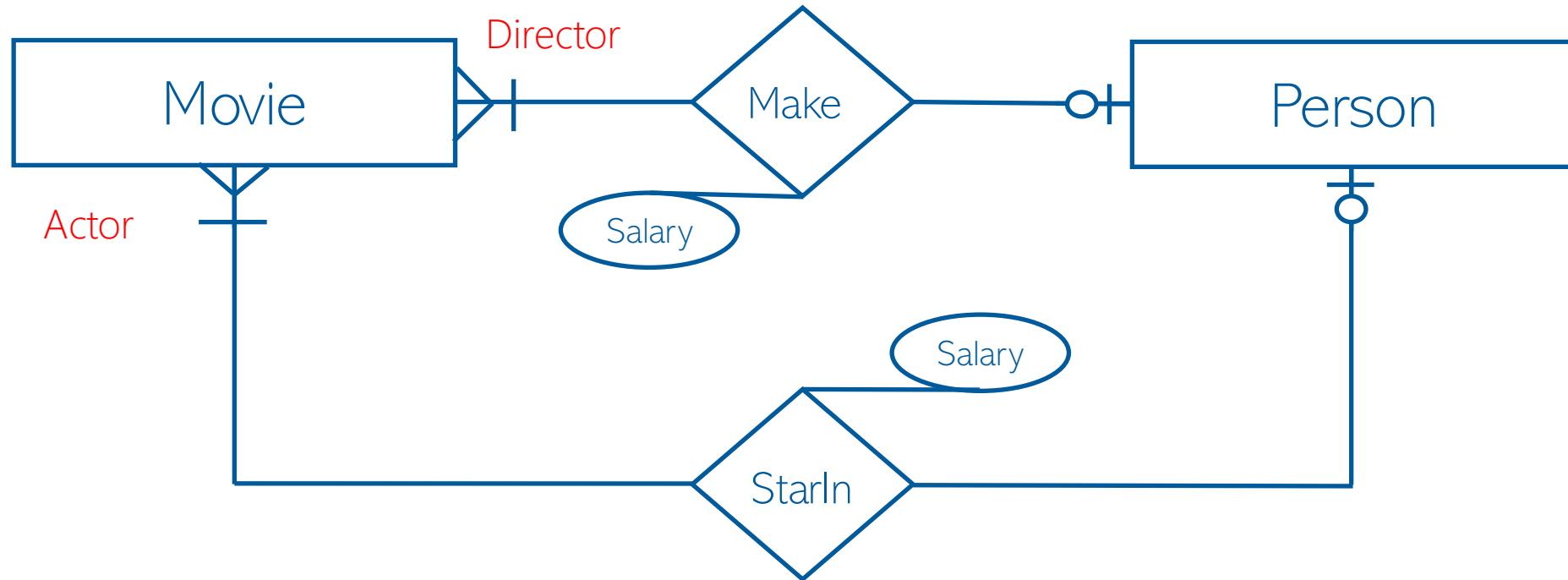
A Producer **must at least** Produce 1 Movie. A Movie can have 0 or 1 Producer.

A Distributor **must at least** Distribute 1 Movie. A Movie can have 0 or 1 Producer.

41

Relationship Set (R): Multiple

A pair of entities of different sets have different types of relationship

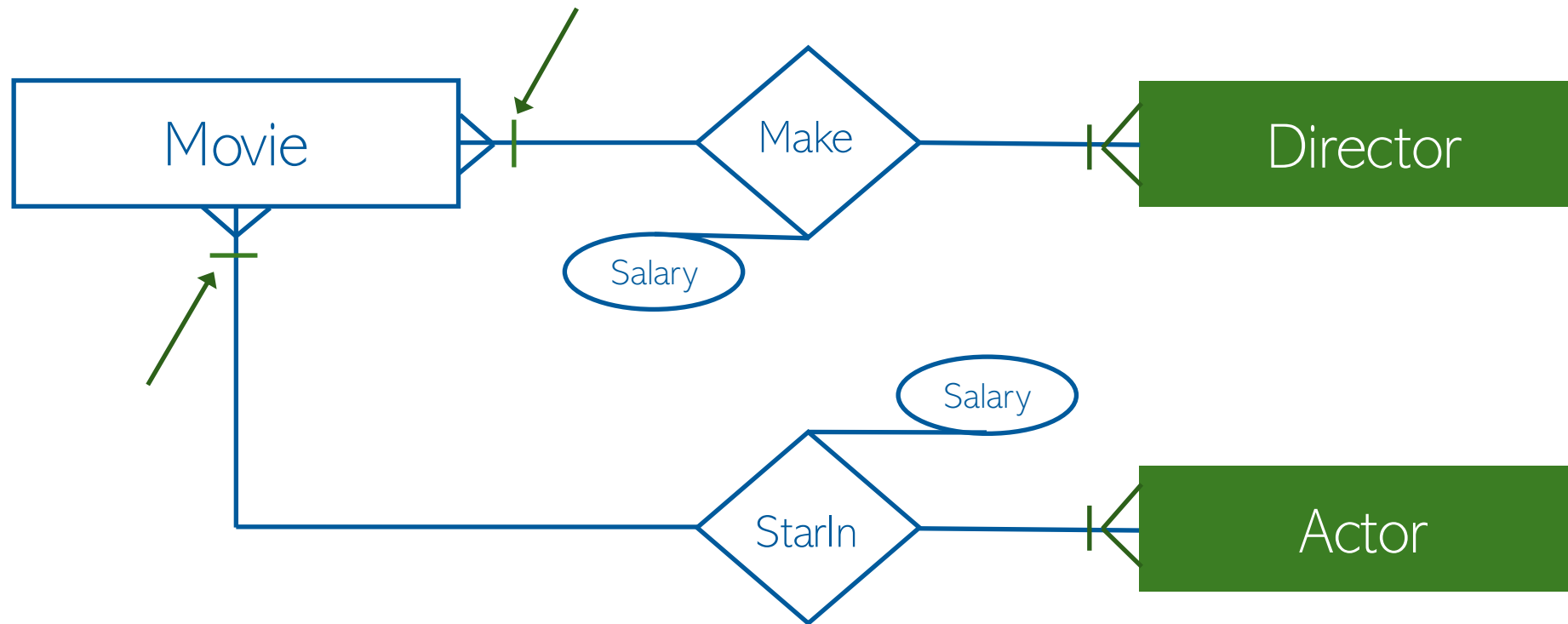


A Person entity is both actor and director.

42

Relationship Set (R): Multiple

A pair of entities of different sets have different types of relationship

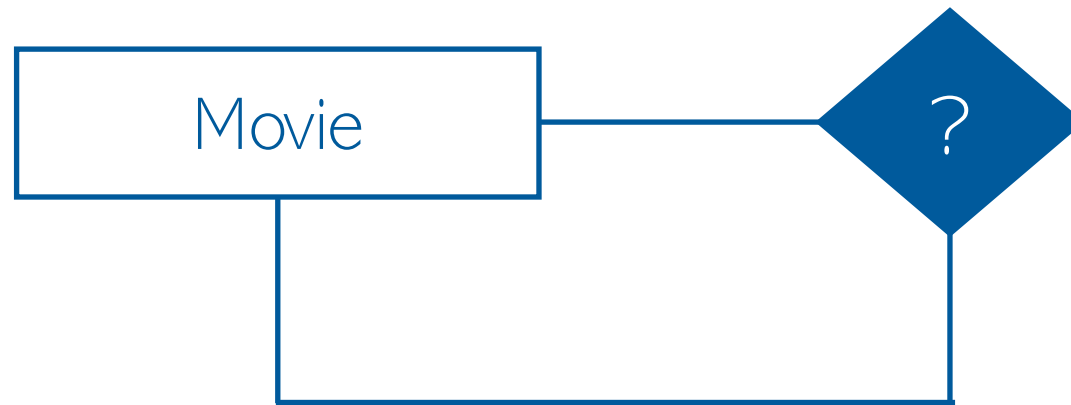


A Director **must at least** Make 1 Movie. A Movie can have many Director.
An Actor **must at least** StarIn 1 Movie. A Movie can have many Actor.

43

Relationship Set (R): Self (Unary)

A pair of entities of the same entity set have relationship

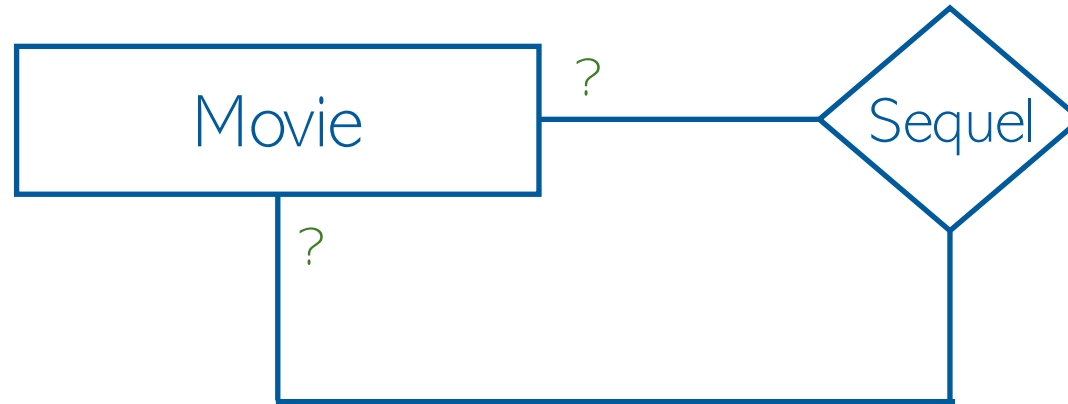


Can a Movie entity participate in a relationship with another Movie entity?

44

Relationship Set (R): Self (Unary)

A pair of entities of the same entity set have relationship



A Movie entity is the original of another Movie entity.

A Movie entity is the sequel of another Movie entity.



45

Relationship Set (R): Self (Unary): Role

A pair of entities of the same entity set have relationship



A Movie entity is an OriginalOf a sequel Movie entity.



46

Relationship Set (R): Self (Unary): Participation (Ordinality)

A pair of entities of the same entity set have relationship



A Movie entity **may not have** a sequel Movie entity.



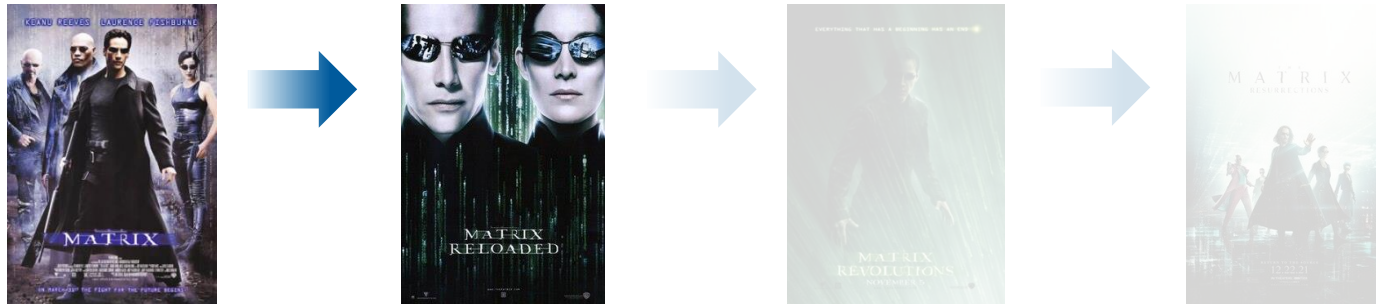
47

Relationship Set (R): Self (Unary): Multiplicity (Cardinality)

A pair of entities of the same entity set have relationship



A Movie entity can be an OriginalOf 1 Movie entity at max.



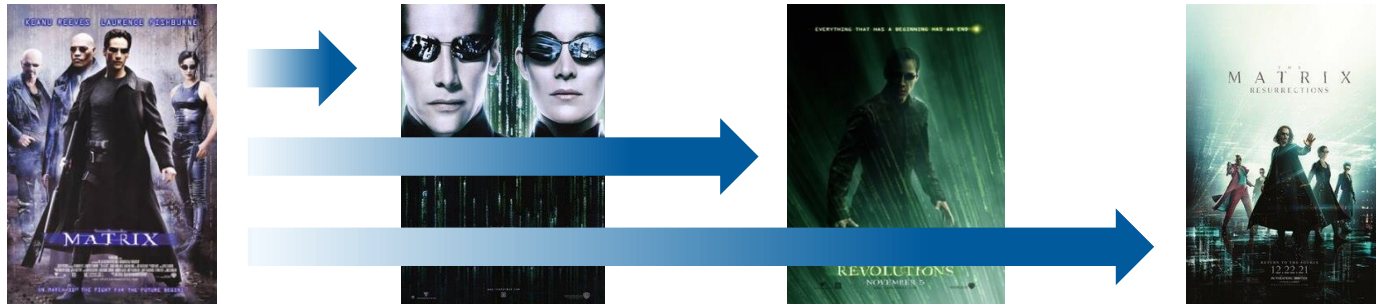
48

Relationship Set (R): Self (Unary): Multiplicity (Cardinality)

A pair of entities of the same entity set have relationship



A Movie entity can be an OriginalOf 1 Movie entity at max.



49

Relationship Set (R): Self (Unary): Role

A pair of entities of the same entity set have relationship



A Movie entity is a SequelOf a Movie entity.



50

Relationship Set (R): Self (Unary): Participation (Ordinality)

A pair of entities of the same entity set have relationship



A Movie entity can be SequelOf no Movie entity.



51

Relationship Set (R): Self (Unary): Multiplicity (Cardinality)

A pair of entities of the same entity set have relationship



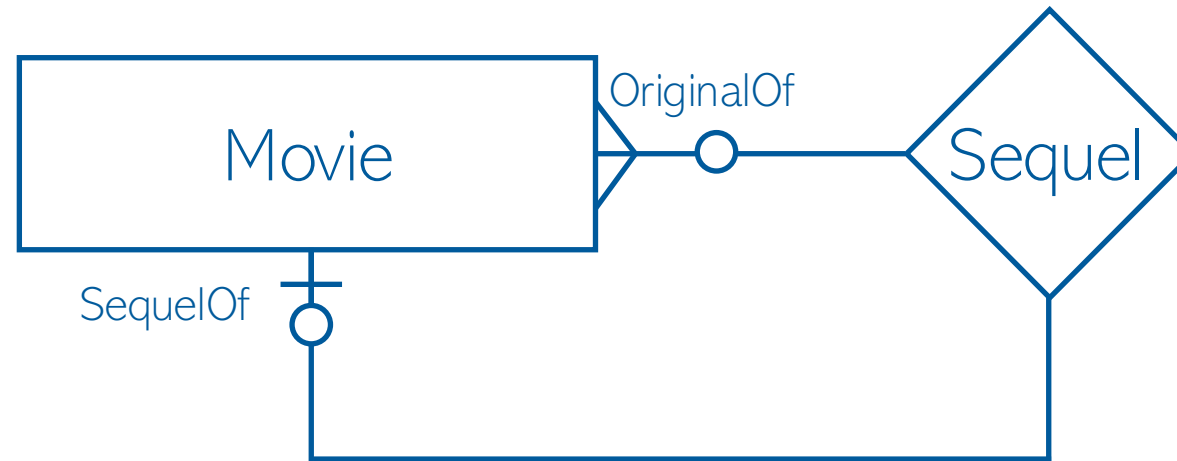
A Movie entity can be SequelOf 1 Movie entity at max.



52

Relationship Set (R): Self (Unary)

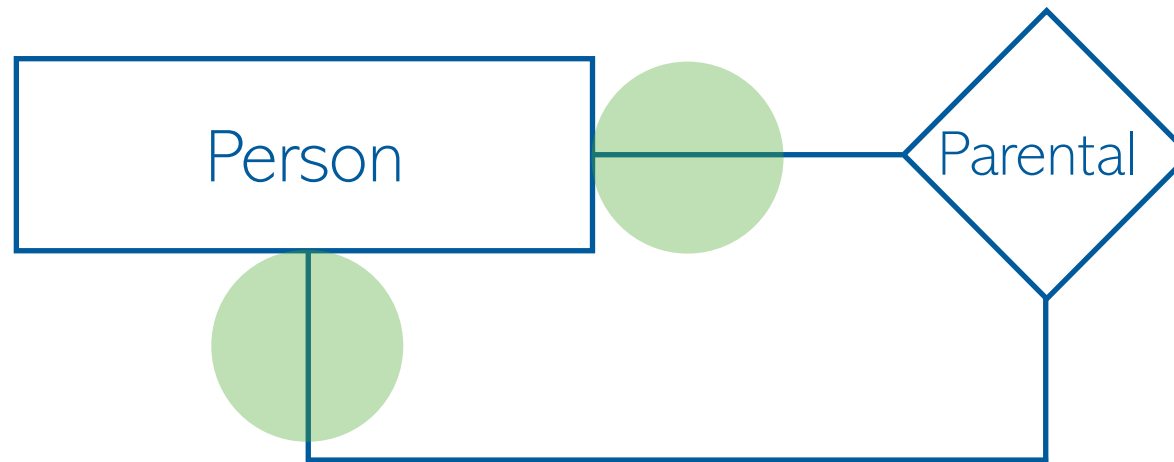
A pair of entities of the same entity set have relationship



53

Relationship Set (R): Self (Unary)

A pair of entities of the same entity set have relationship



54

Relationship Set (R): Self (Unary)

A pair of entities of the same entity set have relationship



55

Relationship Set (R): Self (Unary)

A pair of entities of the same entity set have relationship



A person must have 1 and only 1 parent.

In some culture, only father. In some culture, can be choose, mother vs. father.



A person must have 1 and only 1 parent.

But for some persons, we do not know the information yet. Missing parent.



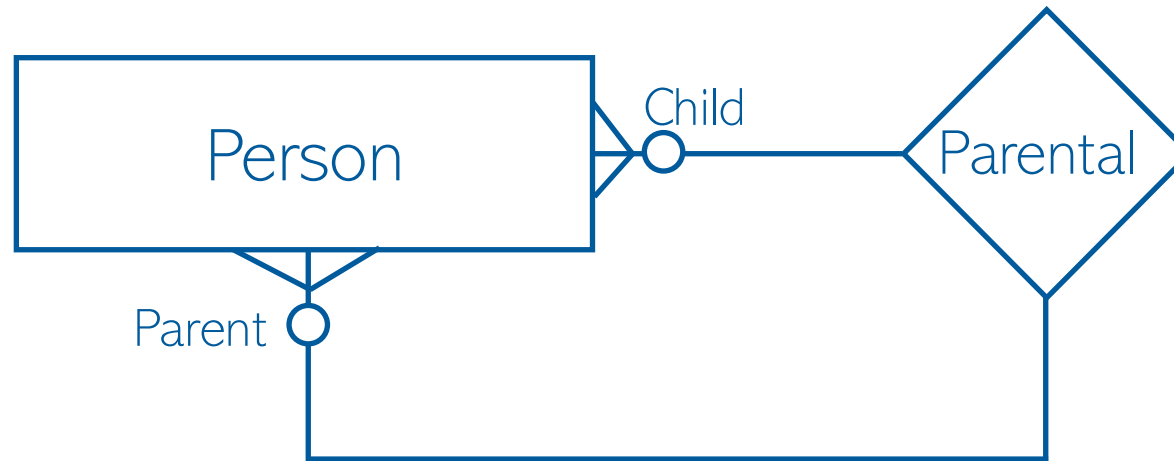
A person must have at least 1 parent

But we want to include both father and mother.

56

Relationship Set (R): Self (Unary)

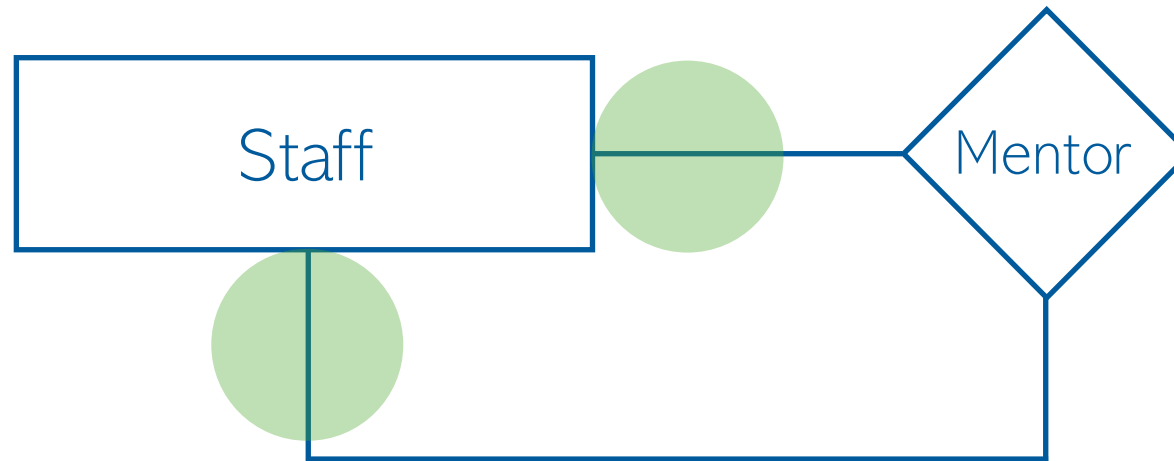
A pair of entities of the same entity set have relationship



57

Relationship Set (R): Self (Unary)

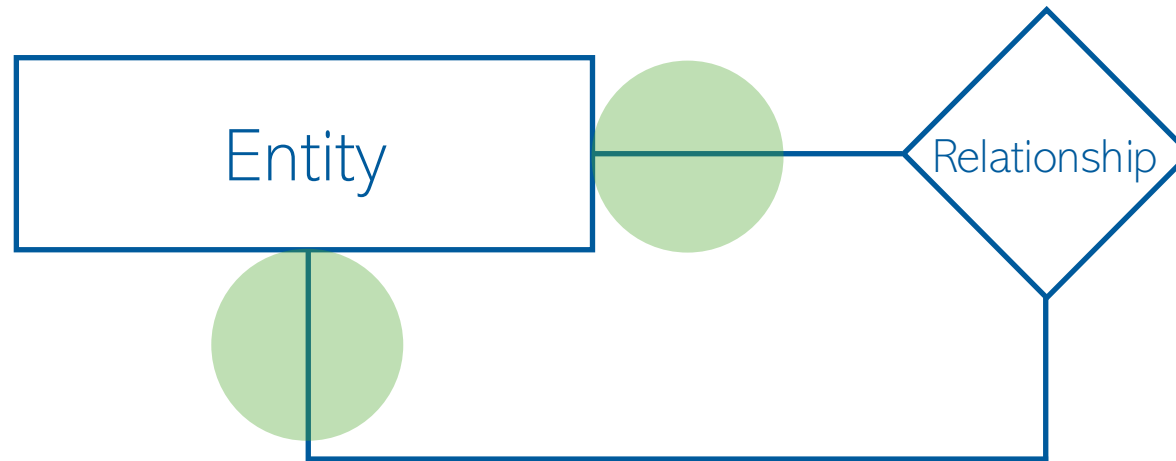
A pair of entities of the same entity set have relationship



58

Relationship Set (R): Self (Unary)

A pair of entities of the same entity set have relationship



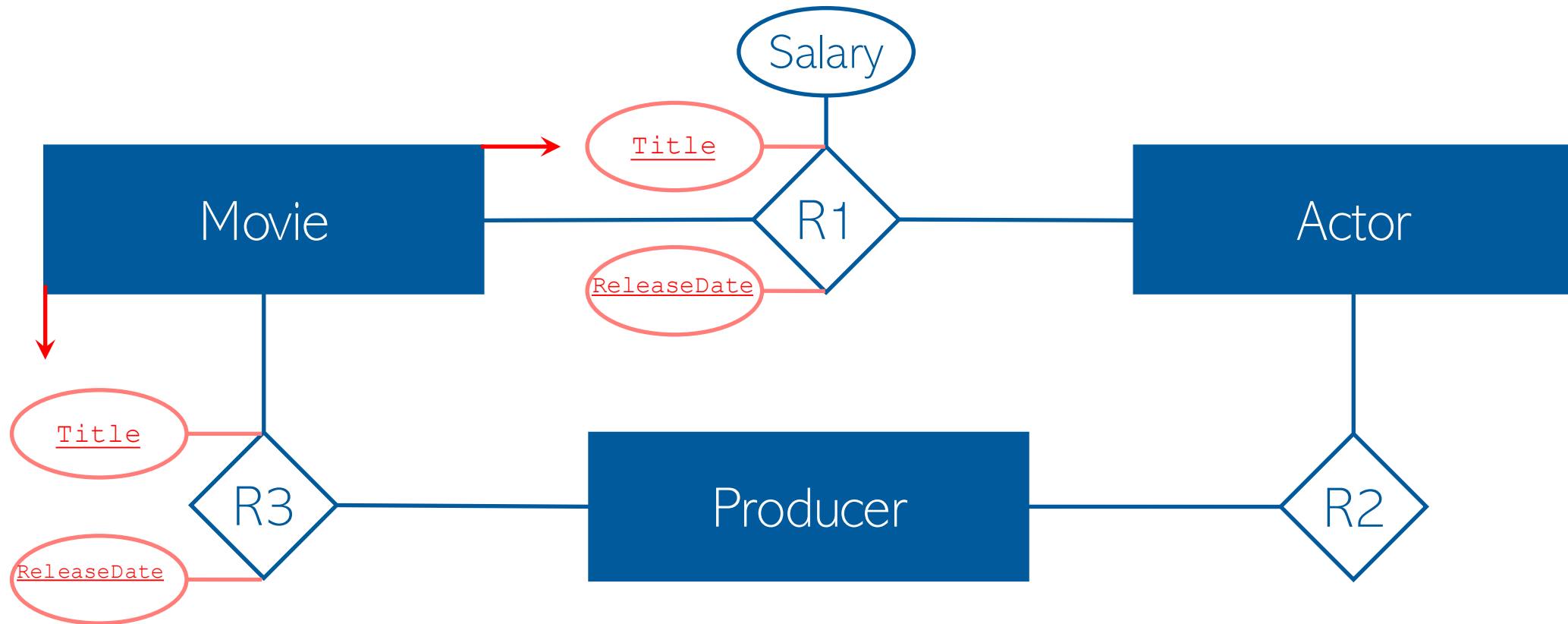
What's the pattern?

Hierarchical relationships among entities of the same entity set.

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Relationship Set (R): Circular; Poor Design

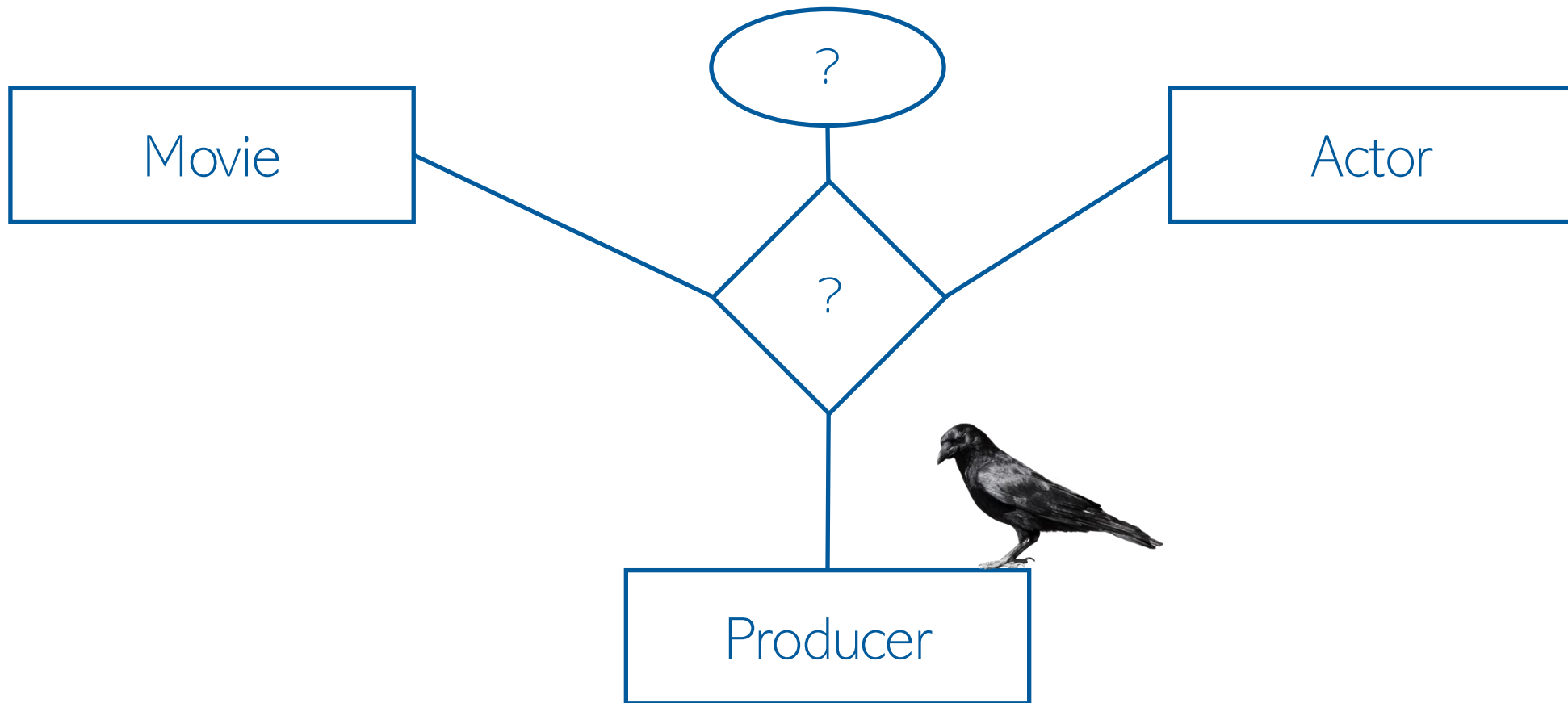
We become biased toward binary relationships ...



60

Relationship Set (R): Ternary

A pair of entities of the same entity set have relationship

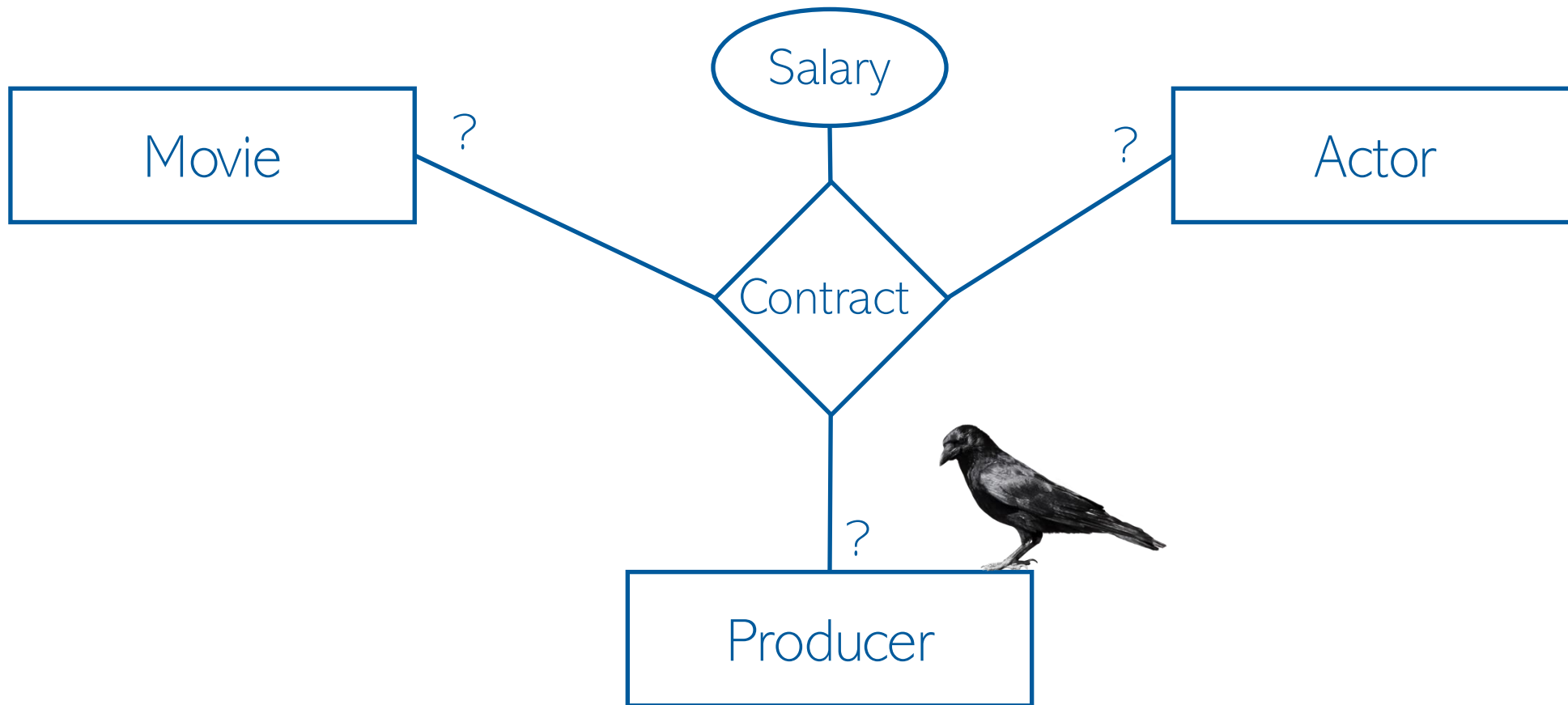


ERDPlus does not support non-binary relationship set/
Either convert to binary, or other tools like www.draw.io

61

Relationship Set (R): Ternary

A pair of entities of the same entity set have relationship

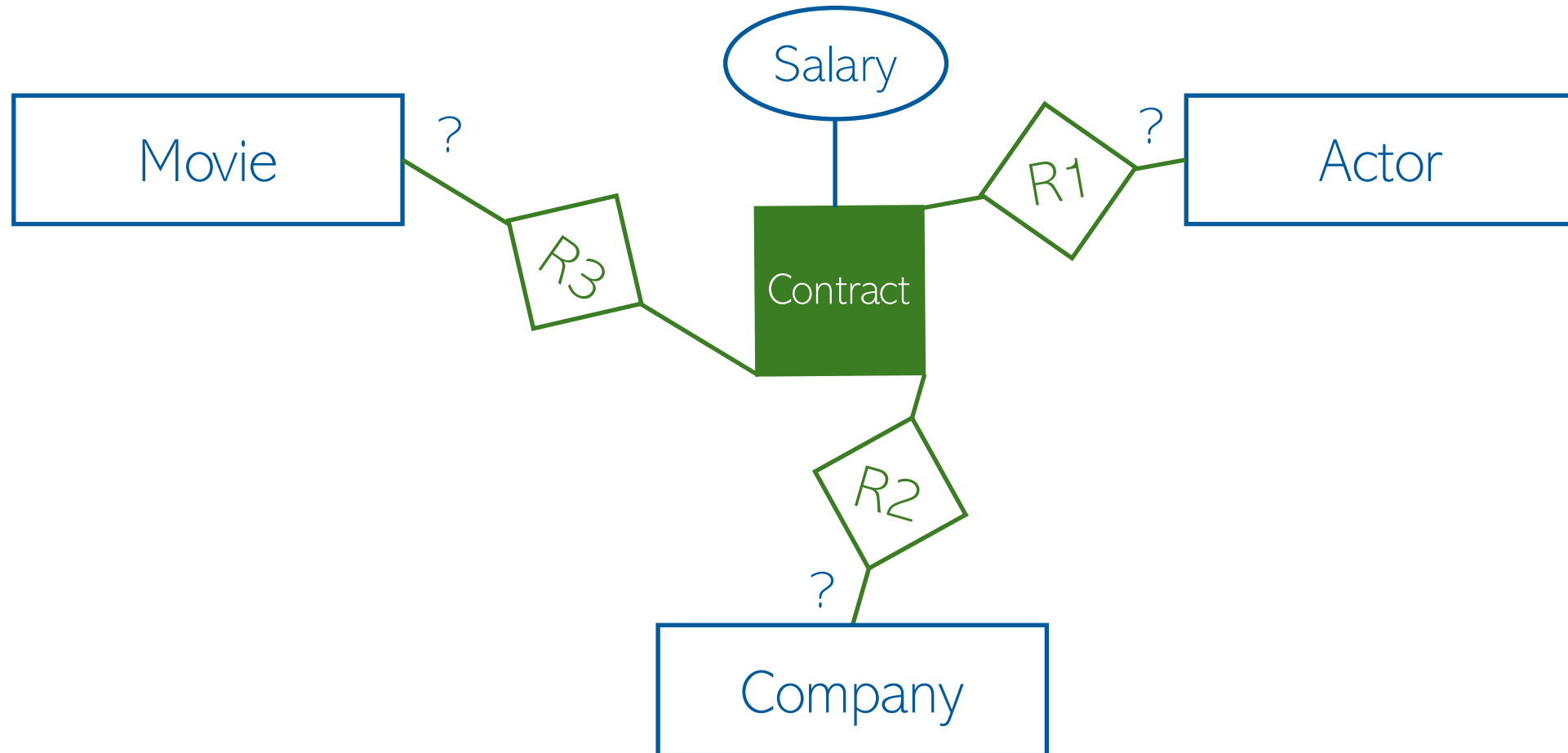


ERDPlus does not support non-binary relationship set/
Either convert to binary, or other tools like www.draw.io

62

Relationship Set (R): Ternary $\rightarrow$ Three Binary

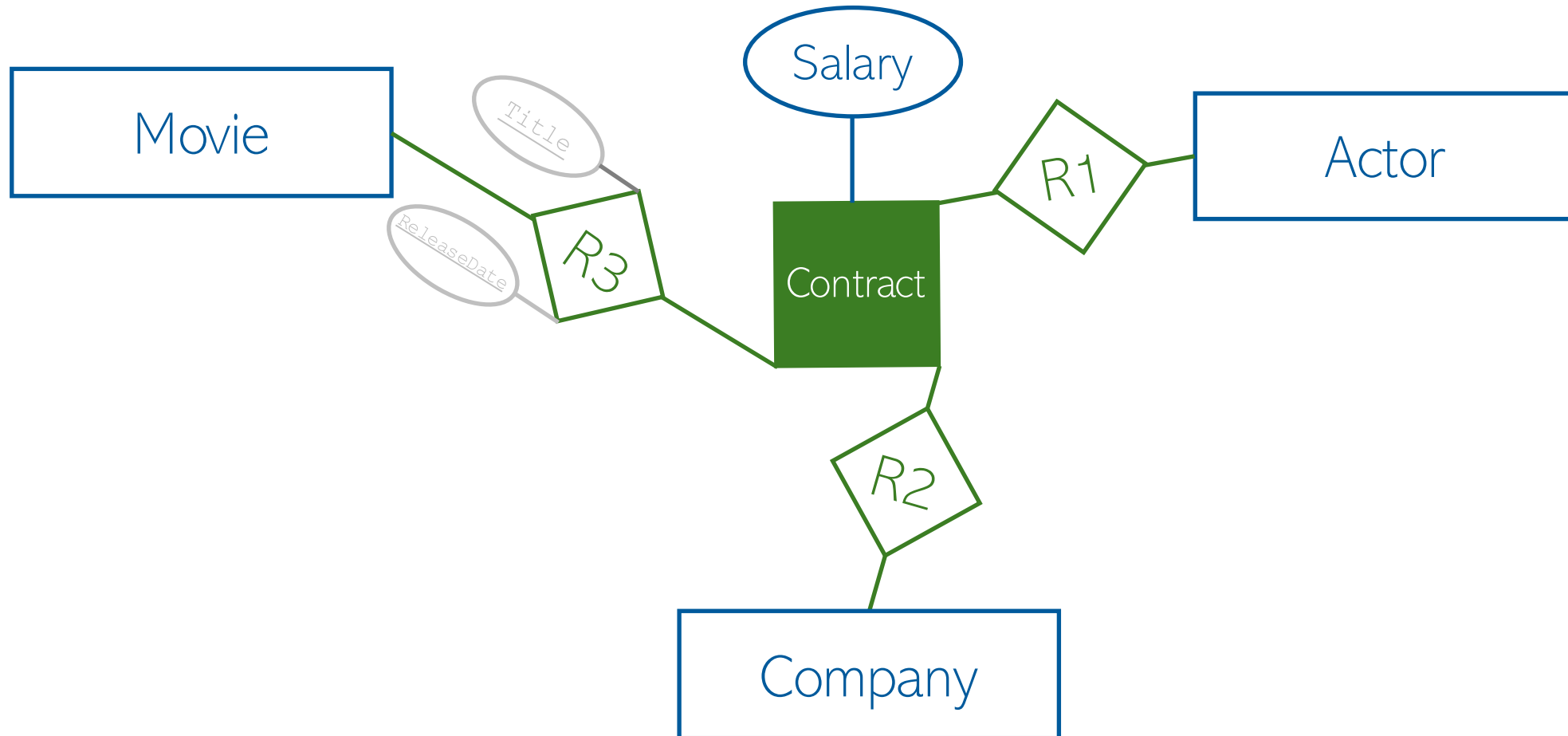
Relationship Set $\rightarrow$ Connecting Entity Set



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Relationship Set (R): Ternary $\rightarrow$ Three Binary

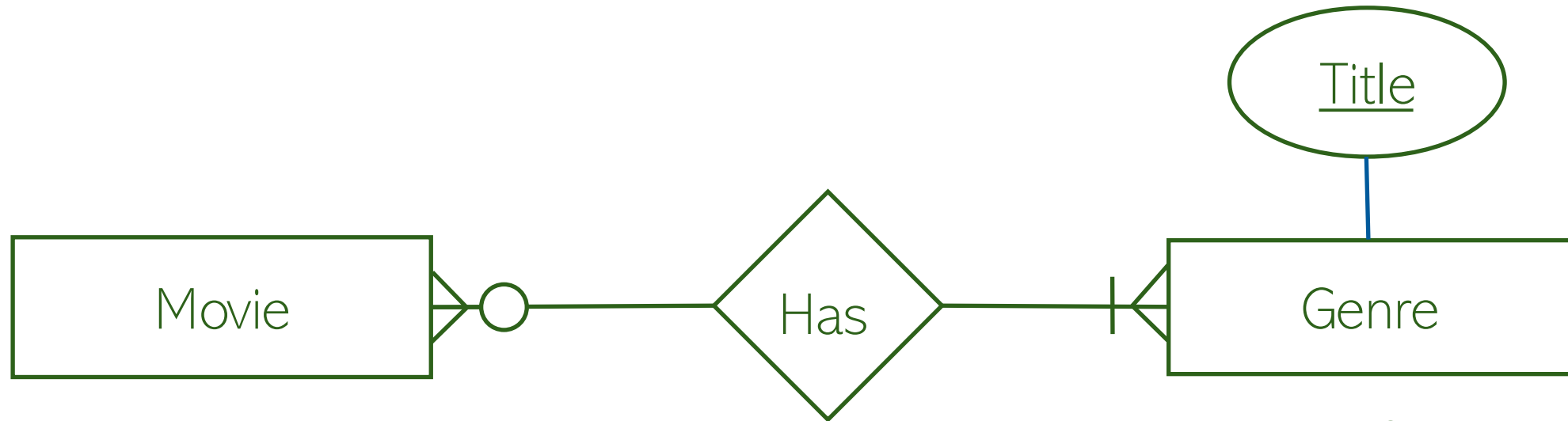
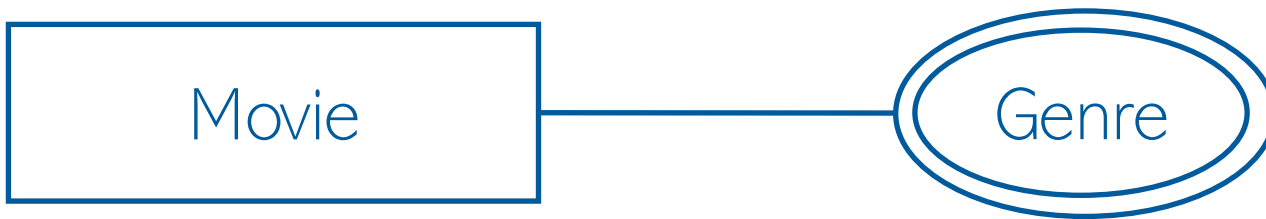
Relationship Set $\rightarrow$ Connecting Entity Set



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Design Questions

Multivalued Attribute vs. “Has” Relationship

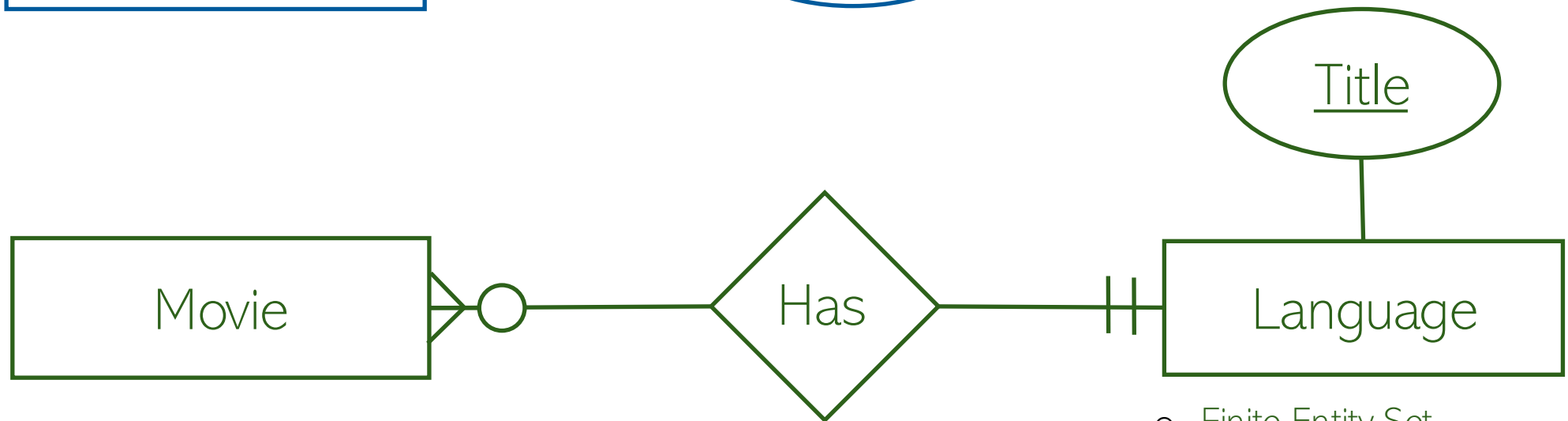
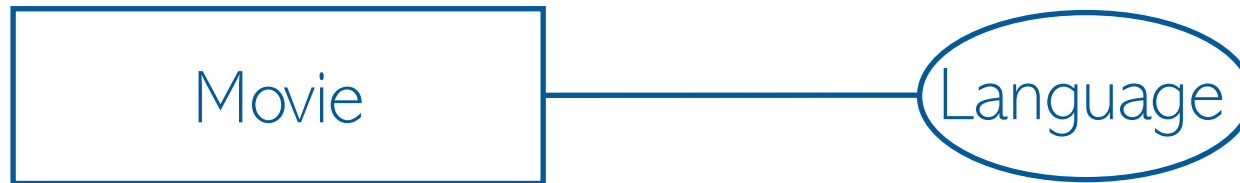


- Finite Entity Set
- Static
- No Multivalued Attribute

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Design Questions

Multivalued Attribute vs. “Has” Relationship

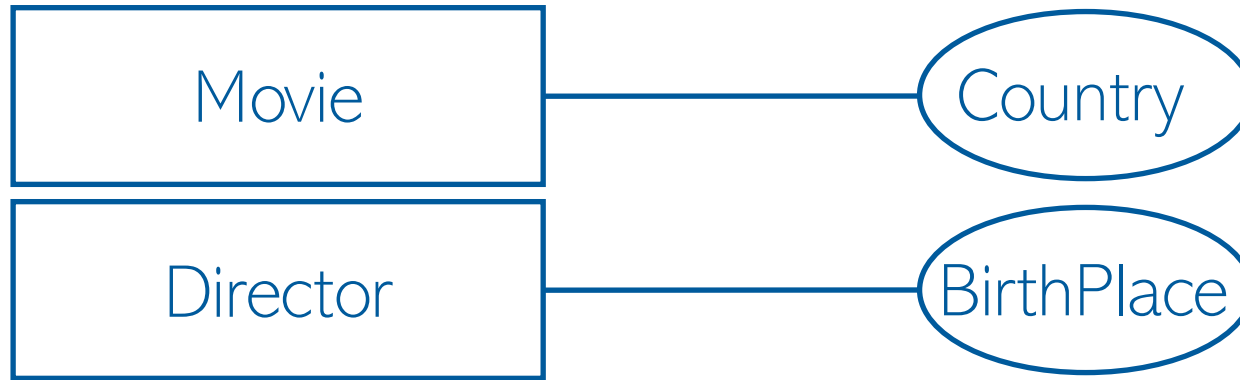


- Finite Entity Set
- Static
- Used in part of the system

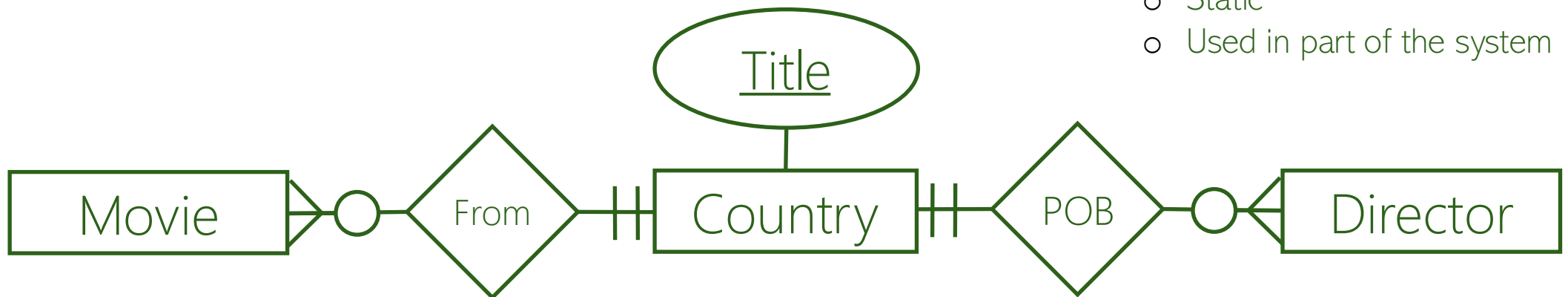
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Design Questions

Multivalued Attribute vs. “Has” Relationship



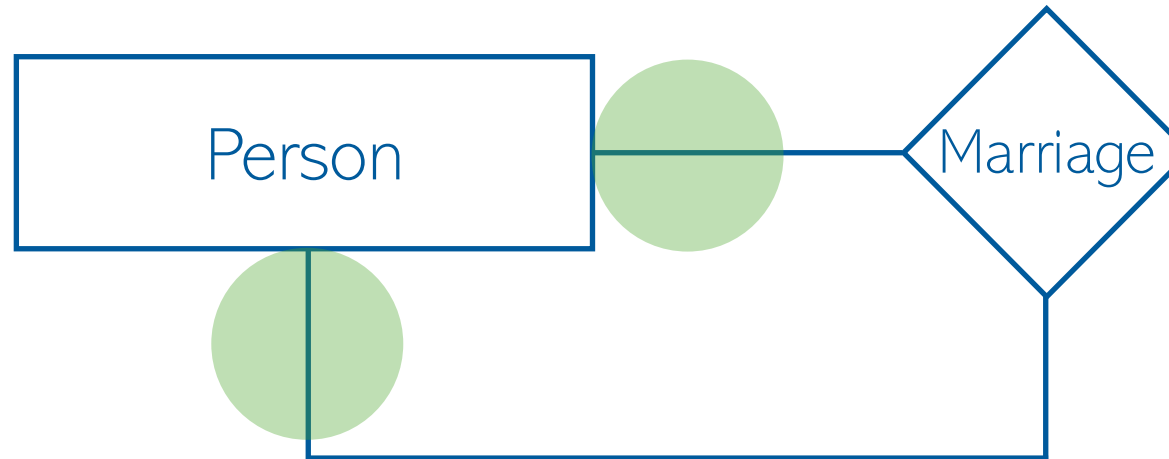
- Finite Entity Set
- Static
- Used in part of the system



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Relationship Set (R): Practice at Home ...

A pair of entities of the same entity set have relationship



Sologamy *'suh-log-uh-mee'* or *'sah-luh-guh-mee'* (autogamy) is marriage by a person to themselves

Polygyny *'puh·li·juh·nee'* a marriage where one man has more than one wife **at the same time**

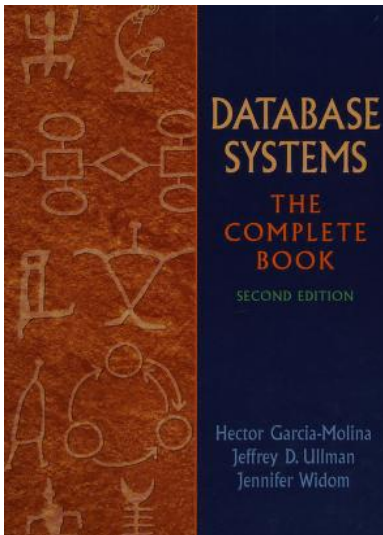
Polyandry *'paa·lee·an·dree'* a form of polygamy where a woman has two or more husbands **at the same time**

Polygamy *'puh·li·guh·mee'* is the practice of being married to more than one spouse **at the same time**

Monogamy *'muh·naa·guh·mee'* is the practice having only one partner at a time

Same Gender/Sex Marriage

Outline: W02B



CH04 (2nd Ed.)

I) Entity

II) Attribute

III) Relationship

III) Extended ER (EER)



Conceptual Level | Extended Entity-Relationship Model